# Local Langlands for $\mathbf{GL}_2(\mathbf{F})$

— ***EYNTKA*** **Series** —

Nathanael Chwojko-Srawley

November 20, 2025

# Contents

#### 0.0.1 Inspiration

Very heavily inspired by [2]

For a review of $p$-adic integers and $p$-adic fields (more generally of local fields), see [5, chapter 1.1].

I want here a rough outline of the Langlands program, in particular that $\mathrm{Gal}(\overline{K}/K)$ corresponds to complex representations of $\mathrm{GL}_n(\mathbb{Q}_p)$, and we want complex representations because this now would give us automorphic forms

# 1

# Smooth Representations

Recall that if $\mathrm{Gal}(K^{\mathrm{ab}}/K)$ is the maximal abelian extension of $K$, it is isomorphic to the profinite completion of the idele group $C_K = \mathbb{I}_K/K^*$. When $K = \mathbb{Q}$, this becomes the restricted product $\prod_p' \mathbb{Z}_p^*$ showing us that there are natural maps

$$\mathrm{Gal}(K^{\mathrm{ab}}/K) \to \mathbb{Q}_p^* = \mathrm{GL}_1(\mathbb{Q}_p) \qquad \text{and} \qquad \mathrm{Gal}(K^{\mathrm{ab}}/K) \to \mathbb{C}^* = \mathrm{GL}_1(\mathbb{C})$$

i.e. one-dimensional representations (over $p$-adic fields or $\mathbb{C}$). When $\mathrm{Gal}(L/K)$ is non-abelian, we get higher-dimensional. In this chapter we shall specifically focus on *complex representation*, as these shall are the object in the Langlands correspondence. In ref:HERE, we shall consider $p$-adic representations.

We start with looking at locally profinite groups in general $\mathbb{Q}_p$ and the multitude of natural objects formed by it are locally profinite. We then look into the right representation of locally profinite groups, in particular we shall require the right notion of smooth representation (as $\mathbb{Q}_p$ is far from being a smooth manifold) that parallels the process for the representation of Lie groups and has the same nice harmonic-analysis properties.

finish explaining here.

## 1.1 Locally Profinite Groups

All finite groups will by assumption have the discrete topology. For a deeper review of topological groups, see [4, chapter 8].

The field $\mathbb{Q}_p$ is a *locally profinite groups* with its topology endowed by the metric $|-|_p$. This is a substantially different topology to $\mathbb{C}$, and hence a moment must be taken to understand better its structure.

**Definition 1.1.1: P-adic Topology**

Let $\mathbb{Q}_p$ be a p-adic field for a (finite) prime $p$ and let $\mathbb{Z}_p$ be the corresponding ring of integers. Define the neighbourhood basis:

$$\mathbb{Z}_p \supseteq p\mathbb{Z}_p \supseteq p^2\mathbb{Z}_p \supseteq \cdots$$

Giving $\mathbb{Z}_p$ a $p$-adic topology which can be easily extended to $\mathbb{Q}_p$. As each $p^k\mathbb{Z}_p$ is a closed ball, this matches the metric topology induced by $|-|_p$:

$$B_r(a) = \{x \in \mathbb{Q}_p \ : \ |x-a|_p < r\}$$

which extends to $\mathbb{Q}_p$ by defining:

$$\left|\frac{m}{n}\right|_p = \frac{|m|_p}{|n|_p}$$

Let us establish some basic properties of this field.

**Lemma 1.1.2: P-adic Field Totally Disconnected**

The field $\mathbb{Q}_p$ with the topology induced by the metric $|-|_p$ is totally disconnected. More generally any space $X$ whose topology is induced by an ultra-metric is totally disconnected.

***Proof*:**
Let $a, b \in \mathbb{Q}_p$. As $a \neq b$, there must exist a $d$ such that:

$$0 < |a-b|_p = p^{-k} = d$$

Let:

$$B_d(a) = \{x \in \mathbb{Q}_p \ : \ |x-a| < d\}$$

By construction, $a \in B_d(a)$ and $b \notin B_d(a)$.

Let us show that

$$B_d(a)^c = \{x \in \mathbb{Q}_p \ : \ |x-a| \geq d\}$$

is also open. Take $x \in B_d(a)^c$ so that $D = |a-x|_p \geq d$ and consider $B_d(x)$ and take $y \in B_d(x)$ so that $\delta = |x-y|_p \leq d$. We want to show that $y \in B_d(a)^c$ so $|y-a|_p \geq d$. Then by the isosceles property of the ultra-metric, for any three points $a, x, y$ two of $d(a,x)$, $d(a,y)$ or $d(x,y)$ must be to the larger one. As $|a-x|_p \geq d$, $|y-a| \geq d$, and so $y \in B_d(a)^c$, showing that $B_d(x) \subseteq B_d(a)^c$ showing that it is *also* open.

Thus, $\mathbb{Q}_p$ is totally disconnected, as we sought to show.

**Lemma 1.1.3: P-adic Integers Open Subgroup**

The subgroup $\mathbb{Z}_p \subseteq \mathbb{Q}_p$ is an open subgroup with the topology induced by the valuation. Hence, it is also closed

This differs strongly from $\mathbb{C}$: if $H \subseteq \mathbb{C}$ is an open subgroup then as noted earlier it must also be closed, hence connected, and hence $H = \mathbb{C}$. As is $\mathbb{Q}_p$ is totally disconnected, it is possible to have clopen subsets that are not the entire set.

***Proof*:**
First, $\mathbb{Z}_p = \{x \in \mathbb{Q}_p \ : \ |x|_p \leq 1\}$. Take $a \in \mathbb{Z}_p$ (so that $|a| \leq 1$) and consider:

$$B_1(a) = \{x \in \mathbb{Q}_p \ : \ |x - a|_p < 1\}$$

I claim $x \in \mathbb{Z}_p$. Indeed:

$$|x|_p = |x + (-a + a)|_p \leq \max(|x - a|_p, |a|_p) = \max(t, 1) = 1$$

where $t < 1$ as $|x - a|_p < 1$. But then $x \in \mathbb{Z}_p$, and hence $B_1(a) \subseteq \mathbb{Z}_p$.

Thus, as $\mathbb{Q}_p$ is a topological group, $\mathbb{Z}_p$ is also closed.

There shall be instances where we work with other similar groups constructed from $p$-adic fields, for example finite extension of $\mathbb{Q}_p$, the ring of integers of $L_\mathfrak{p}/\mathbb{Q}_p$, $\overline{\mathbb{Q}}_p$, $\mathbb{C}_p$, $\mathrm{GL}_n(\mathbb{Q}_p)$ or $\mathrm{GL}_n(\mathcal{O}_{L/\mathbb{Q}_p})$ (or matrix subgroups of this group), $\mathrm{GL}_n(\mathfrak{m}_\mathfrak{p})$ and so forth. Thus, it is worth going over a general theory that encapsulates the topology of these spaces. Let us first recall some basic properties of topological groups:

- The topology of $G$ is translation invariant: If $U \subseteq G$ is open, then $Ux$ and $xU$ is open
- If $H \leqslant G$, $G/H$ has the quotient topology, and if $H \trianglelefteq G$ then $G/H$ is a topological group.
- If $K_1, K_2$ are compact subset of $G$, so is $K_1 K_2$
- subgroups in relation to open/closed:
  - if $H \leqslant G$ is open, it must be closed ($\bigcup_{g \in G \setminus H} gH$ is the union of open sets and the compliment of $H$). More strictly, if $H$ is open if and only if it contains an open neighbourhood around the identity (if $e \in U \subseteq H$ with open $U$, then $hU \subseteq H$ and $H = \bigcup_{h \in H} hU$). If $G$ is connected, it has no non-trivial proper open subgroups. We shall frequently be working with totally disconnected groups.
  - If $H \leqslant G$ is an open subgroup, then the quotient space $G/H$ and $H \backslash G$ each have the discrete topology.
  - A closed subgroup $H \leqslant G$ need not be open (ex. $\mu_n \subseteq \mathbb{C}$). However if $H \leqslant G$ is closed and finite index, it is open (the finite union of closed is closed). If $G$ is first-countable, $H$ is closed if it contains all its limit points (more generally, it it contains all the limits of nets).
  - A subgroup need not be open or closed (ex. $\mathbb{Q} \subseteq \mathbb{R}$).
  - if $H$ is a subgroup of $G$, so it $\overline{H}$ (the closure under limits/nets).
  - IF $G_0 \subseteq G$ is the connected component of the identity, then $G_0$ is closed, $G_0$ is the intersection of all normal subgroups of $G$, and $G/G_0$ is the largest totally disconnected quotient.
- symmetric neighbourhood:

- For every open neighborhood of $e$, say for example $U$, there is a symmetric neighborhood (if $A \subseteq G$ such that $A = A^{-1}$, we'll say $A$ is *symmetric*) $V$ of $e$ such that $V \subseteq U$
- For every neighborhood $U$ of $e$, there is a neighborhood $V$ of $e$ with $VV \subseteq U$

**Lemma 1.1.4: Characterizing Hausdorffness**

Let $G$ be a topological group, $e \in G$ the identity, and $H = \bigcap_{\substack{U \subseteq G \text{ open} \\ e \in U}} U$ Then:

1. $H$ is a subgroup
2. $H$ is the closure of $e$, i.e. $\overline{e} = H$
3. $G/H$ is Hausdorff
4. $G$ is Hausdorff if and only if $H = \{e\}$

***Proof*:**
exercise

Henceforth, we shall always assume Hausdorffness unless specified otherwise.

**Definition 1.1.5: Profinite Group**

Take a diagram finite groups with the discrete topology. Then a *profinite group* is a topological group that is the colimit of the system. Given any topological group $G$, we can take its *profinite completion* by taking:

$$\widehat{G} = \varprojlim_N G/N$$

where $N$ ranges over all finite index open normal subgroups ordered by containment.

Any group $G$ can be given the profinite topology by taking the cosets of finite-index normal subgroups as a basis. The reader should check that if $G$ is a topological group and $\widehat{G}$ is its profinite completion, then $\varphi : G \to \widehat{G}$ is a continuous morphism and $G$ is dense in $\widehat{G}$.

**Example 1.1.6: Profinite Groups**

We have the following "trivial" examples:

1. Let $G$ be finite. Then all normal subgroups have finite index and there are finitely many. It follows that $G \cong \widehat{G}$.
2. Take $G = \mathbb{Q}$. Then $\mathbb{Q}$ has only one finite-index normal subgroup, namely itself. Hence $\widehat{\mathbb{Q}} = \{1\}$.

The following is the "usual" example and is better suited for the intuition of profinite groups. Let $G = \mathbb{Z}$. Then

$$\widehat{\mathbb{Z}} = \varprojlim_n \mathbb{Z}/n\mathbb{Z} \cong \prod_p \mathbb{Z}_p$$

where $\mathbb{Z}_p = \varprojlim_n \mathbb{Z}/p^n\mathbb{Z}$ is the $p$-adic completion of $\mathbb{Z}$. The main idea is that for any element $n \in \mathbb{Z}$, in chain of normal subgroups (say $p^k\mathbb{Z} \subseteq p^{k+1}\mathbb{Z} \subseteq \cdots$) the element mod $p^k\mathbb{Z}$ eventually stabilize.

For example, take $p = 7$ and $n = 129'654 = 2 \cdot 3^3 \cdot 7^4$. Then the value is zero for the until $\mathbb{Z}/7^5\mathbb{Z}$ where it becomes $12'005 \mod 7^n$ for $n \in \{5, 6\}$ and then $129'654$ forever onward. $\mathbb{Z}_p$ allows for never-stabilizing chains (mirroring how stable sequences of rational numbers converge to rational numbers).

For Langlands, the important non-abelian example is the profinite completion of galois Galois groups, namely $\mathrm{Gal}_K(\overline{K})$. See [6] or a quick overview or [1] for more details.

More generally, profinite groups can be seen as the object that captures all the finite behaviour "up to infinity" (or philosophically, all the "potential infinity") of a system. In the $\mathbb{Z}_p$ example, it appeared when taking a solution mod each $p^n$ and for galois it comes from amalgamating all symmetries of roots of some infinite family of (finite) polynomial. Another place this shall show up is in the definition of the étale fundamental group, see ref:HERE.

The above examples show that the map $\varphi : G \to \widehat{G}$ could be injective, surjective, both.

**Lemma 1.1.7: Pro-finite Completion Injective**

Let $G$ be a topological group and $\widehat{G}$ it's profinite completion. Then $G \to \widehat{G}$ is injective if the intersection all finite-index open normal subgroups of $G$ is trivial

***Proof*:**
exercise.

The completion comes equipped with a natural universal property:

**Theorem 1.1.8: Universal Property of Profinite Completion**

let $\varphi : G \to H$ be continuous homomorphism with $H$ a profinite group. Then there is a unique continuous homomorphism such that the following diagram commutes:

$$\begin{tikzcd} G \arrow[r, "\phi"] \arrow[dr, "\varphi"'] & \widehat{G} \arrow[d, "\exists!"] \\ & H \end{tikzcd}$$

***Proof*:**
exercise (hint: recall $G$ is dense in $\widehat{G}$).

A striking result is that all profinite groups can be characterized by a simple topological property:

**Theorem 1.1.9: Characterizing Profinite Groups**

A topological group is profinite if and only if it is a totally disconnected compact group

***Proof*:**
Let $G$ be profinite. Then it is a closed subgroup of a direct product of finite groups, hence compact and totally disconnected.

Conversely, suppose $G$ is totally disconnected and compact. We must show it comes from an inverse system of finite groups. will finish later.

**Corollary 1.1.10: Distinguishing Profinite Completion**

Let $G$ be a profinite group. Then $G$ is naturally isomorphic to its profinite completion, or even stronger:
$$G \cong \varprojlim G/U$$
where $U$ ranges over all *open* normal subgroups ordered by contain need.
If $G$ is isomorphic to its profinite completion *as a group* (i.e. is strongly complete) if and only if every finite index subgroup is open

***Proof*:**
exercise.

**Definition 1.1.11: Locally Profinite Group**

A topological group $G$ is *locally profinite* if it is Hausdorff and every open neighbourhood of the identity contains a compact open subgroup of $G$.

Notice how there is no mention of an inverse system. The reader should show that the definition imples $G$ is totally disconnected, and hence if it is also compact then we recover a profinite group. Every discrete group is finite, closed subgroups of a locally profinite group are locally profinite, and so are quotients by closed normal subgroups. A locally profinite subgroup is locally compact. It is an exercise to show that any locally compact totally disconnected group is locally profinite. The canonical example for these notes of a locally profinite group is the non-Archimedean local field $(F, +)$ as proven in [5]. Similarly, $(F^\times, \times)$ is a profinite group by passing through $e^x$, or:

$$U_F^n = 1 + \mathfrak{p}^n$$

Being profinite is closed under products, hence $F^n$ is profinite, and $(M_n(F), +)$ is locally profinite under addition. The group $\mathrm{GL}_n(F) \subseteq M_n(F)$ under multiplication is an open topological subgroup. It is locally profinite as $\mathrm{GL}_n(\mathcal{O}_F)$ is profinite (it is compact as it is homeomorphic to a closed subset of $\mathcal{O}_F^{n^2}$ which is a product of compact groups, and it is open as $\mathcal{O}_F = \{x \in F \ : \ |x|_\mathfrak{p} \leq 1\}$ is an open set. Note that this was all done in coordinates, more generally this applies to $\mathrm{End}_F(V)$ and $\mathrm{Aut}_F(V)$.

Let us now consider character's on locally profinite groups:

**Theorem 1.1.12: Continuous Character**

let $\varphi : G \to \mathbb{C}^\times$ be a group homomorphism where $G$ is locally profinite. Then the following are equivalent:

1. $\varphi$ is continuous
2. the kernel of $\varphi$ is open

If $\varphi$ satisfies one of these conditions and $G$ is the union of its compact open subgroups (such as $\mathbb{Q}_p$), then $\mathrm{im}\varphi \subseteq S^1 \subseteq \mathbb{C}^\times$.

***Proof*:**
If the kernel is open, then if $U \subseteq \mathbb{C}^\times$ and $K = \ker \varphi$,

$$\varphi^{-1}(U) = \bigcup_{g \in \varphi^{-1}(U)} gK$$

showing $\varphi$ is continuous. Conversely, Let $\varphi$ be continuous and $U$ an open neighbourhood of 1. Then $\varphi^{-1}(U)$ is open, and thus by local-profiniteness contains a compact open subgroup $K \subseteq G$. As this applies to any $U$, we can pick $U$ sufficiently small so that it contains no non-trivial subgroup of $\mathbb{C}^\times$. As the image of group is a group and $K \subseteq \varphi^{-1}(U)$, $K \subseteq \ker \varphi$. As $K$ is open, $\ker \varphi$ contains an open subgroup and hence by topological group theory $\ker \varphi$ is open.

Next, $S^1$ is the unique maximal compact subgroup of $\mathbb{C}^\times$. If $K'$ is a compact subgroup of $G$, $\varphi(K') \subseteq S^1$, giving the 2nd result as we sought to show.

This condition has interesting algebraic consequences, for example take the map $\mathbb{Z} \to S^1$ by sending 1 to $e^{2\pi i\theta}$ where $\theta$ is irrational. As $\mathbb{Z}$ is dense in $\mathbb{Z}_p$, this map extends to a map $\mathbb{Z}_p \to S^1$. It's kernel is $\{0\}$, i.e. it is not open, and so this map is not continuous with the profinite topology on $\mathbb{Z}_p$.

**Corollary 1.1.13: Continuous Maps To General Linear Group**

Let $\varphi : G \to \mathrm{GL}_n(\mathbb{C})$ be a group homomorphism where $G$ is locally profinite. Then $\varphi$ is continuous if and only if the kernel of $\varphi$ is open.

If $G$ is the union of compact open subgroup, the maps into $U(n)$ I'll prove this later.

***Proof*:**
Note that $\mathrm{GL}_n(\mathbb{C})$ is a lie group and that there exists an open neighbourhood of $I_n$ that contains no non-trivial subgroups, the argument follows.

**Definition 1.1.14: Continuous Character**

A *character* for a locally profinite group $G$ is a continuous homomorphism

$$\varphi : G \to \mathbb{C}^\times$$

We shall often write $1_G$ or just 1 for the trivial character (i.e. the constant character).

If $\varphi(G) \subseteq S^1$, $\varphi$ is called a *unitary character*.

summary of next section On page 12 in [2], he defines $\widehat{F}$ to be the set of all characters of $F$. As $F$ is a local field, it is the union of its compact open subgroup $\mathfrak{p}^n$ for $n \in \mathbb{N}$, hence all characters are unitary. Then if $\varphi \neq 1$ is a character, there exists $d \in \mathbb{Z}$ such that $\mathfrak{p}^d \subseteq \ker(\varphi)$; the least such number is defined as the level of $\varphi$ .Fixing a $d$, the set of characters of level $\leq d$ is a subgroup. This defines a duality which extends to the multiplicative set, in other words this is indeed the right notion of character!

## 1.2 Smooth Representations of Locally Profinite Groups

All representations shall be complex unless specified otherwise. When working with $\mathbb{Q}_p$ representations and more generally representations of locally profinite groups, it is important to take into account the topology, or else we will get representations that give no valuable information:

**Example 1.2.1: "Useless" Representation**

Take the algebraic closure of $\mathbb{Q}_p$, $\overline{\mathbb{Q}}_p$. By field theory, it embeds into $\mathbb{C}$ (using the axiom of choice). Thus there is an injective ring homomorphism $\mathbb{Q}_p \to \mathbb{C}$ and hence a (valid) representation:

$$\pi : \mathrm{GL}_n(\mathbb{Q}_p) \to \mathrm{GL}_n(\mathbb{C})$$

But this has disregarded all topological (and hence arithmetic) information of $\mathbb{Q}_p$.

The first step to fix this would be to only consider continuous representations as is done in the representation of topological groups. Often when $G$ is not compact, we get a vector space $V$ that is infinite dimensional with which is either a Banach or Hilbert space and $\mathrm{GL}(V)$ is the set of all invertible bounded linear operators; this thread is further explored in [3].

When restricting to locally profinite groups, there is a useful stronger notion called *smooth*. It must be that this notion of smoothness is different from smoothness for manifolds as we will be looking at maps $\mathbb{Q}_p \to \mathbb{C}$ which don't share the local topology ($\mathbb{Q}_p$ is also totally disconnected, so there is no way to put a reasonable atlas on it). Instead, the intuition comes from how smoothness (or differentiability) means that zooming in close will get the same linear, or "constant" if we change coordinates in our perspective. In terms of locally profinite groups, as they are the union of open compact subgroups, a representation being constant on these groups abstractly shares this same property. This loose analogy turns out to be very practical, hence:

**Definition 1.2.2: Smooth Map For Locally Profinite Groups**

Let $G$ be a locally profinite group. Then a map $f : G \to \mathbb{C}^n$ is *smooth* if it is locally constant, namely for any $x \in G$ there exists an open neighbourhood $U$ containing $x$ such that $f(x) = f(y)$ for all $x, y \in U$.

**1.2.3 Precision in Vocabulary Via Harmonic Analysis** There is a more precise comparison when considering the Fourier transform with distributions, so we'll assume [4] for the following.

1. On $\mathbb{R}$: For the Fourier transform, the space of "ideal" functions is the Schwartz space, consisting of functions that are both smooth ($C^\infty$) and rapidly decaying. The Fourier transform is a perfect map on this space: the transform of a smooth, rapidly decaying function is another smooth, rapidly decaying function.

2. On $\mathbb{Q}_p$: There is an analogous theory of Fourier analysis. The "ideal" space of functions (called the Schwartz-Bruhat space) consists of functions that are locally constant and have compact support. The Fourier transform on $\mathbb{Q}_p$ is a good map on this space: the transform of a locally constant, compactly supported function is another locally constant, compactly supported function. See this link for more details.

**Proposition 1.2.4: Equivalence Notion of Smoothness**

Let $G$ be a locally profinite group and $f : G \to \mathbb{C}^n$ a map. Then $f$ is smooth if and only if for *any* subset $U \subseteq \mathbb{C}^n$ (which is *not* necessarily open), $f^{-1}(U) \subseteq G$ is open.

Thus, if $f$ is smooth, it is continuous with $\mathbb{C}^n$ having the discrete topology.

***Proof*:**
exercise.

The key in the notion of smoothness being useful is that on $\mathbb{Q}_p$ there are non-trivial locally constant functions:

**Example 1.2.5: Smooth Functions On Locally Profinite Groups**

1. The $p$-adic valuation $|-|_p : \mathbb{Q}_p^\times \to \mathbb{R}$ is smooth. To see this, take any $x \in \mathbb{Q}_p$ so that $|x|_p = p^{-n}$ for some $n \in \mathbb{Z}$ giving the range. Then the pre-image of $p^{-n}$ is $p^n\mathbb{Z}_p^\times$ which is open.

   Note that this is not smooth on $\mathbb{Q}_p$ because the pre-image of 0 is 0, i.e. there is no open neighbourhood of 0 on which the absolute value is constant. This can even be stretched to the absolute value on $\mathbb{R}$, when restricting to $\mathbb{R}^\times$ it is smooth, but naturally it is not smooth at 0.

2. The map $\mathrm{GL}_n(\mathbb{Q}_p) \to \mathbb{R}$ given by $g \mapsto |\det(g)|_p$ is smooth.

Often, a smooth function should be thought of as being well-defined modulo cosets of an open subgroup.

By identifying $M_n(\mathbb{C}) \cong \mathbb{C}^{n^2}$, we get the notion of a smooth representation:

**Definition 1.2.6: Smooth Representation**

Let $G$ be a locally profinite group. Then a *complex representation* (or just *representation*) of $G$ is a pair $(\pi, V)$ where $V$ is a complex vector space and $\pi : G \to \mathrm{Aut}_{\mathbb{C}}(V)$ a group homomorphism.

A representation $(\pi, V)$ is called *smooth* if for every $x \in V$, there is a compact open subgroup $K_x \subseteq G$ such that for all $k \in K_x$:

$$\pi(k) \cdot x = x$$

Equivalently, if $V^K$ denotes the space of $\pi(K)$-fixed vector in $V$,

$$V = \bigcup_K V^K$$

where $K$ ranges over all compact open subgroups of $G$.

A smooth representation is called *admissible* if $V^K$ is finite dimensional for all compact open $K \subseteq G$.

**Proposition 1.2.7: Smooth Representation Alternative**

A representation $\pi : G \to \mathrm{Aut}_{\mathbb{C}}(V)$ is smooth if

$$\mathrm{Stab}_G(v) = \{g \in G \ : \ \pi(g) \cdot v = v\}$$

is an open subgroup of $G$ for all $v \in V$

***Proof*:**
exercise.

If $(\pi, V)$ is a smooth representation of $G$, then any $G$-stable subspace of $V$ is another smooth representation, and similarly if $U \subseteq V$ is a $G$-subspace, the induced representation of $G$ on $V/U$ is smooth. A representation $(\pi, V)$ is *irreducible* if if $V \neq 0$ and $V$ has no nontrivial $G$-stable subspaces. A $G$-homomorphism between two representations $(\pi_1, V_1)$, $(\pi_2, V_2)$ is a map that commutes with the group action induce by $\pi_i$.

**Example 1.2.8: Smooth Representations**

1. If $(\mathbb{C}, \chi)$ is a one-dimensional representation, by theorem 1.1.12 it is smooth if and only if it is continuous. Hence in one dimensions these two concepts are the same. Indeed, if $\chi(g) \cdot v = v$, as $\chi(g)$ is a complex number and $v$ is nonzero we have:

$$\chi(g) = 1$$

Hence, $\chi$ is smooth if and only if the kernel is open (and as $G$ is locally profinite it contains a compact subgroup), which by theorem 1.1.12 is true if and only if $\chi$ is continuous.

By corollary 1.1.13, any finite dimensional continuous representation is smooth.

2. Let $G = \mathbb{Q}_p^*$; let us look for maps $\psi : \mathbb{Q}_p^* \to \mathbb{C}^*$ which as $\mathbb{Q}_p$ is the union of compact open sets by theorem 1.1.12 we know every maps into $S^1$. As $\mathbb{Q}_p^* \cong \mathbb{Z} \times \mathbb{Z}_p^*$ (since $x = p^n u$ for $n \in \mathbb{Z}$ and $u \in \mathbb{Z}_p^*$; namely $n = \nu_p(x)$), and $\psi$ is a group homomorphism, the map is determined by where it maps $(1, 0)$ and $\mathbb{Z}_p^*$. Let us look at the two parts seperatly:

$$\psi((c, u)) = \psi((c, 1)) \cdot \psi((0, u)) = \psi_1(c) \cdot \psi_2(u)$$

   Thus, for any choice of $s \in \mathbb{C}$, let:

$$\psi_1(\nu_p(x)) = p^{-s\nu_p(x)}$$

   Next, lets look at the units. Note that this includes $1 \in \mathbb{Z}_p^*$, and we know that the kernel must be open, and as $\varphi(1) = 1$, that means that an open neighbourhood around $1 \in \mathbb{Z}_p^*$ maps to $1 \in \mathbb{C}^*$. The open neighborhoods around 1 are:

$$U_k = 1 + p^k \mathbb{Z}_p = \left\{u \in \mathbb{Z}_p^* \ : \ u \equiv 1 \bmod (p^k)\right\}$$

   Thus, for some $k \in \mathbb{N}$ $\psi_2(U_k) = 1$ By the isomorphism theorem the cosets will be:

$$\mathbb{Z}_p * /U_K \cong (\mathbb{Z}/p^k\mathbb{Z})^*$$

   But now $\psi_2$ reduces to $(\mathbb{Z}/p^k\mathbb{Z})^* \to \mathbb{C}^*$ which is just the familiar roots of unity maps. Note how this is "ramified". For this reason, $\psi_1$ is often called the *unramified component* while $\psi_2$ is called the *ramified component.*

   Combining these together, if we let $\psi_1(x) = |x|_p^s$ and $\psi_2(x) = \psi(x)$:

$$\chi(x) = |x|_p^s \psi(x)$$

   These classify all continuous/smooth characters of $\mathbb{Q}_p^*$.

3. Let us next consider the next dimension up: $G = \mathrm{GL}_n(\mathbb{Q}_p)$ an $V = \mathbb{C}$, that is smooth characters on $\mathrm{GL}_n(\mathbb{Q}_p)$. Then As $\mathbb{C}^*$ is commutative, $\chi : \mathrm{GL}_n(\mathbb{Q}_p) \to \mathbb{C}^*$ must be trivial on the commutator. It is an exercise in linear algebra to show that $[G, G] = \mathrm{SL}_n(\mathbb{Q}_p)$[a]. As the identity has determinant 1, $I \in \mathrm{SL}_n(\mathbb{Q}_p)$ and as $\chi$ is a group homomorphism it must map to 1. Thus $\chi(\mathrm{SL}_n(\mathbb{Q}_p)) = 1$.

   Next, for any $A, B \in \mathrm{GL}_n(\mathbb{Q}_p)$ such that $\det(A) = \det(B)$ so that $AB^{-1} \in \mathrm{SL}_n(\mathbb{Q}_p)$, we must have $\chi(A) = \chi(B)$ as $\chi$ is a homomorphism. But then, $\chi$ is only dependent on the determinant of a matrix.

   From this, to can construct a character $\chi : \mathrm{GL}_n(\mathbb{Q}_p) \to \mathbb{C}^*$, we choose a 1-dimensional character $\psi : \mathbb{Q}_p^* \to \mathbb{C}^*$ and compose it with the determinant map $A \mapsto \det(A)$ so that:

$$\chi(A) = \psi(\det(A))$$

   It can be shown that all characters are of this form.

4. Let us consider $G = \mathbb{Q}_p$ and $V = \mathbb{C}$ so that we are looking at characters $\chi : \mathbb{Q}_p \to \mathbb{C}^*$. These are all formed by the *standard character*. For any $x \in \mathbb{Q}_p$ the $p$-adic representation is of the form

$$x = \sum_n^\infty a_i p^i \qquad n \in \mathbb{Z},\ a_i \in \{0, 1, ..., p-1\}$$

The $p$-adic fractional part is the negative indexed summands; let $\{x\}_p$ represent this. Define:

$$\psi_p : \mathbb{Q}_p \to \mathbb{C}^* \qquad \psi_p(x) = e^{2\pi i \{x\}_p}$$

Notice the kernel of this map is $\mathbb{Z}_p$, and hence it is continuous/smooth.

5. We can form characters on $\mathfrak{m}_p$ by restricting characters on $\mathbb{Q}_p$. Show that $\hat{\mathfrak{m}}_p \cong \mathbb{Q}_p/\mathbb{Z}_p$.

6. For a continuous non-smooth example, let $G = \mathbb{Z}_p$ and $V = C(\mathbb{Z}_p, \mathbb{C})$ be the $\mathbb{C}$-vector space of continuous characters with the uniform-convergence topology. Then take the left-regular representation $(\rho, V)$:
$$(\rho(g) \cdot f)(x) = f(x - g)$$
The reader can show this is continuous. Let us look at the stabilizer of the vector $g \mapsto |g|_p$. Note this map is certainly continuous as the norm-map is continuous, and thus $| - |_p \in C(\mathbb{Z}_p, \mathbb{C})$. Let us consider the stabilizer of $| - |_p$ that is all, $g \in \mathbb{Z}_p$ such that :
$$g \cdot | \bullet |_p = | \bullet - g|_p = | \bullet |_p$$
Let $g \in \mathrm{Stab}_{\mathbb{Z}_p}(| - |_p)$ so that:
$$|g - x|_p = |x|_p \forall x \in \mathbb{Z}_p$$
If $x \neq g$, then it easy to see that equality *fails*, hence the only stabilizer is 0. But then $\mathrm{Stab}_{\mathbb{Z}_p}(| - |_p) = \{0\}$ which is *not* open.

7. (principal series representation) Let:
$$B = \left\{ \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \in \mathrm{GL}_2(\mathbb{Q}_p)\} \right\}$$
Let us define a character on $B$ by choosing two characters $\chi_1, \chi_2 : \mathbb{Q}_p^* \to \mathbb{C}^*$ given by:
$$\varphi\left( \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \right) = \chi_1(a)\chi_2(c)$$
Now, define the complex vector space $V$ to be smooth complex function $f : \mathrm{GL}_2(\mathbb{Q}_p) \to \mathbb{C}$ such that:
$$f(bg) = \psi(b) f(g) \qquad b \in B,\ g \in \mathrm{GL}_2(\mathbb{Q}_p)$$
Then define the action to be:
$$(\rho(h) \cdot f)(g) = f(gh)$$
Then this is an infinite dimensional smooth representation; for a choice of $\chi_1, \chi_2$ denote it $I(\chi_1, \chi_2)$. It is almost always irreducible, but there shall be there are non-reducible representations given the right choices of $\chi_1, \chi_2$, see example 1.2.13.

By Schur's Lemma, the abelian groups have no higher dimensional irreducible representations. For higher finite dimensional smooth representations of the non-abelian groups, ref:HERE shall show that they are *all* a direct sum of 1 dimensional representations, even though these groups are not abelian! This shows that to capture some of their more non-abelian structure of these groups, it will be important to look at (well-behaved) infinite dimensional representations, hence the need to add the smooth condition.

[a] this is true as long as the field $F$ is not the field with two elements

**Proposition 1.2.9: Generalized Maschke's Theorem**

Let $G$ be compact (thus profinite) and $(\pi, V)$ an irreducible representation of $G$. Then $V$ must be finite-dimensional.

***Proof*:**
If $0 \neq v \in V$, then $v \in V^K$ for an open subgroup $K \subseteq G$. As $G$ is profinite, $[G : K]$ is finite and so:

$$\{\pi(g) \cdot v \ : \ g \in G/K\}$$

spans $V$. We can then take the smaller $K' = \bigcup_{g \in G/K} gKg^{-1}$ which is an open normal subgroup of $G$, still of finite index, and the action is invariant on $K'$. Thus, $V$ can be seen as an irreducible representation of the finite discrete group $G/K'$, and hence is certainly finite dimensional.

As there are many hierarchies of types of representations in representation theory (semi-stable, crystalline, etc), we shall purposefully specify the category that has been described thus far:

**Definition 1.2.10: Smooth Representation Category**

Let $G$ be a locally profinite group. Then $\mathrm{Rep}_{\mathrm{smooth},K}(G)$ is the smooth representation category whose objects are smooth representations $(V, \pi)$ and morphisms are $G$-equivariant maps. The representations shall always be smooth, and hence we write $\mathrm{Rep}_K(G)$ instead of $\mathrm{Rep}_{\mathrm{smooth},K}(G)$. The field $K$ is dropped if it is clear from context; by default in this book it shall be $\mathbb{C}$.

By a similar argument as the usual category for representations of finite groups, $\mathrm{Rep}(G)$ is an abelian category. When working with finite groups, we saw that representations can be thought of as $\mathbb{C}[G]$-modules. With the smoothness condition imposed, there are more elements in the $\mathbb{C}[G]$-module $V$ than smooth representations; we shall need to limit to *Hecke algebras* as shall be discussed in section 1.4.

Let us next upgrade the standard decomposition result from finite group representation theory:

**Proposition 1.2.11: Semisimple Equivalence**

Let $G$ be a locally profinite group, and let $(\pi, V)$ be a smooth representation of $G$. The following conditions are equivalent:

1. $V$ is the sum of its irreducible $G$-subspaces;
2. $V$ is the direct sum of a family of irreducible $G$-subspaces;
3. any $G$-subspace of $V$ has a $G$-complement in $V$.

***Proof*:**
(1) $\Rightarrow$ (2). Take the set of irreducible $G$-subspaces $U_i$ of $V$

$$\{U_i : i \in I\}$$

such that $V = \sum_{i \in I} U_i$. Consider the set $\mathcal{I}$ of subsets $J$ of $I$ such that the sum $\sum_{i \in J} U_i$ is direct.

The set $\mathcal{I}$ is non-empty; Indeed, for an induction argument suppose we have a totally ordered set $\{J_a : a \in A\}$ of elements of $\mathcal{I}$. Put $J = \bigcup_{a\in A} J_a$. If the sum $\sum_{j\in J} U_j$ is not direct, there is a finite subset $S$ of $J$ for which $\sum_{j\in S} U_j$ is not direct. Since $S$ must be contained in some $J_a$, this is impossible. Therefore $J \in \mathcal{I}$. We can now apply Zorn's Lemma to get a maximal element $J_0$ of $\mathcal{I}$. For this set, we have

$$V = \bigoplus_{a\in J_0} U_i,$$

as required for (2).

(2) $\Rightarrow$ (3). In (3), let $W$ be a $G$-subspace of $V$. By (2), we can assume that $V = \bigoplus_{i\in I} U_i$, for a family $(U_i)$ of irreducible $G$-subspaces of $V$. We consider the set $\mathcal{J}$ of subsets $J$ of $I$ for which $W \cap \sum_{i\in J} U_i = 0$. Again, the set $\mathcal{J}$ is nonempty and inductively ordered by inclusion. If $J$ is a maximal element of $\mathcal{J}$, the sum $X = W + \bigoplus_{j\in J} U_j$ is direct. If $X \neq V$, there is $i \in I$ with $U_i \not\subset X$, so the sum $X + U_i$ is direct, and $J \cup \{i\} \in \mathcal{J}$, contrary to hypothesis. Thus (2) $\Rightarrow$ (3).

(3) $\Rightarrow$ (1) Let $V_0$ be the sum of all irreducible $G$-subspaces of $V$ and write $V = V_0 \oplus W$, for some $G$-subspace $W$ of $V$. Assume for a contradiction that $W \neq 0$. By its definition, the space $W$ has no irreducible $G$-subspace. However, there is a non-zero $G$-subspace $W_1$ of $W$ which is finitely generated over $G$. By Zorn's Lemma, $W_1$ has a maximal $G$-subspace $W_0$, and then $W_1/W_0$ is irreducible. We have $V = V_0 \oplus W_0 \oplus U$, for some $G$-subspace $U$ of $V$, and hence a $G$-projection $V \to U$. The image of $W_1$ in $U$ is an irreducible $G$-subspace of $U$, hence an irreducible subspace of $V$ which is not contained in $V_0$. This is nonsense, so $V = V_0$ and (3) $\Rightarrow$ (1).

**Definition 1.2.12: Semisimple Representation**

A representation $(\pi, V)$ is *$G$-semisimple* if it satisfies one of the equivalent conditions of proposition 1.2.11.

Unlike finite groups, locally profinite groups will have (many) non-semisimple complex representations:

**Example 1.2.13: Reducible non-semisimple representation**

The details will be left for the reader. Let $\chi_1(x) = |x|_p$ and $\chi_2(x) = |x|_p^{-1}$ and take the principal series representation $I(\chi_1, \chi_2)$. This representation has a unique 1-dimensional subrepresentation which is a character of $\mathrm{GL}_2(\mathbb{Q}_p)$; denote it $\chi$. The quotient of $I(\chi_1, \chi_2)$ by this representation is called the *Steinberg representation* and is denoted St. This fit into a short exact sequence:

$$0 \to \chi \to I(\chi_1, \chi_2) \to \mathrm{St} \to 0$$

However, this sequence *does not split*. Thus, it is *reducible* but *indecomposable*, hence *not semisimple*.

The compact opens subgroups shall fair better:

**Lemma 1.2.14: Compact Open Semistable Representation**

Let $G$ be a locally profinite group, and let $K$ be a compact open subgroup of $G$. Let $(\pi, V)$ be a smooth representation of $G$. The space $V$ is the sum of its irreducible $K$-subspaces:

$$V = \sum_{\substack{W \subseteq V \\ \text{irred. } K\text{-subspace}}} W$$

We say that $V$ is $K$*-semisimple*.

***Proof*:**
Let $v \in V$. As in example 1.2.8, $v$ is fixed by an open normal subgroup $K'$ of $K$, and it generates a finite-dimensional $K$-space $W$ on which $K'$ acts trivially. Thus $W$ is effectively a finite-dimensional representation of the finite group $K/K'$ and so is the sum of its irreducible $K$-subspaces. Since $v \in V$ was chosen at random, the lemma follows.

Thus, the following is a meaningful definition:

**Definition 1.2.15: Isotypic Component**

Let $G$ be a locally profinite group and $K \subseteq G$ a compact open subgroup. Let $\widehat{K}$ denote the set of equivalence classes of irreducible representations. Let $\rho \in \widehat{K}$ and $(\pi, V)$ be a smooth representation of $G$. Then:

$$V^\rho = \sum_{\substack{W \subseteq V \\ K\text{-subspace} \\ \text{of class } \rho}} W$$

is called the $\rho$*-isotypic component* of $V$.

By definition, $V^K$ is the isotypic subspace for the class of trivial representation 1 of $K$.

**Proposition 1.2.16: $K$-isotypic Decomposition**

Let $(\pi, V)$ be a smooth representation of $G$ and let $K$ be a compact open subgroup of $G$.

1. The space $V$ is the direct sum of its $K$-isotypic components:

$$V = \bigoplus_{\rho \in \widehat{K}} V^\rho.$$

2. Let $(\sigma, W)$ be a smooth representation of $G$. For any $G$-homomorphism $f : V \to W$ and $\rho \in \widehat{K}$, we have

$$f(V^\rho) \subset W^\rho \text{ and } W^\rho \cap f(V) = f(V^\rho).$$

***Proof*:**
By proposition 1.2.11:

$$V = \bigoplus_{i \in I} U_i,$$

for a family of irreducible $K$-subspaces $U_i$ of $V$. We let $U(\rho)$ be the sum of those $U_i$ of class $\rho$. We then have

$$V = \bigoplus_{\rho \in \widehat{K}} U(\rho).$$

If $W$ is an irreducible $K$-subspace of $V$ of class $\rho$, then $W \subset U(\rho)$; otherwise, there would be a non-zero $K$-homomorphism $W \to U_i$, for some $U_i$ of some class $\tau \neq \rho$. We deduce that $V^\rho = U(\rho)$ and (1) follows.

In (2), the image of $V^\rho$ is a sum of irreducible $K$-subspaces of $W$, all of class $\rho$ and therefore contained in $W^\rho$. Moreover, $f(V)$ is the sum of the images $f(V^\tau)$, $\tau \in \widehat{K}$, and $f(V^\tau) \subset W^\tau$. Since the sum of the $W^\tau$ is direct, $f(V)$ is the direct sum of the $f(V^\tau)$ and the second assertion follows.

We frequently use part (2) of the Proposition in the following context:

**Corollary 1.2.17: Exactness reduced to compact subgroup**

Let $a : U \to V$, $b : V \to W$ be $G$-homomorphisms between smooth representations $U, V, W$ of $G$. The sequence

$$U \xrightarrow{a} V \xrightarrow{b} W$$

is exact if and only if

$$U^K \xrightarrow{a} V^K \xrightarrow{b} W^K$$

is exact, for every compact open subgroup $K$ of $G$.

***Proof*:**
exercise.

If $H$ is a subgroup of $G$, we define

$$V(H) = \text{the linear span of } \{v - \pi(h)v : v \in V, h \in H\}.$$

In particular, $V(H)$ is an $H$-subspace of $V$.

**Corollary 1.2.18: Splitting Using Compact Subgroups**

Let $G$ be a locally profinite group, and let $(\pi, V)$ be a smooth representation of $G$. Let $K$ be a compact open subgroup of $G$. Then

$$V(K) = \bigoplus_{\substack{\rho \in \hat{K} \\ \rho \neq 1}} V^\rho, \quad V = V^K \oplus V(K),$$

and $V(K)$ is the unique $K$-complement of $V^K$ in $V$.

***Proof*:**
The sum $W = \bigoplus V^\rho$, with $\rho$ not trivial, is a $K$-complement of $V^K$ in $V$. There is therefore a $K$-surjection $V \to V^K$ with kernel $W$. Clearly, $V(K)$ is contained in the kernel of any $K$-homomorphism $V \to V^K$; we conclude that $W$ contains $V(K)$. On the other hand, if $U$ is an irreducible $K$-space of class $\rho \neq 1$, then $U(K) = U$, so $V^\rho \subset V(K)$.

The next important concept is that of induced representation and how we can construct representations using them. Recall that in the finite case a subgroup $H \subseteq G$ which has a representation $(W, \sigma)$ will have a natural induced representation $(\mathbb{C}[G]_{\mathbb{C}[H]}W, \mathrm{Ind}_H^G \sigma)$ where the space $\mathbb{C}[G]_{\mathbb{C}[H]}W$ can be represented as $H$-equivariant functions and $\mathrm{Ind}_H^G$ acts on the set of these functions. The following generalizes this:

**Definition 1.2.19: Induced Representation**

Let $G$ be a locally profinite group, and let $H \subseteq G$ be a closed subgroup of $G$ so that $H$ is also locally profinite. Let $(\sigma, W)$ be a smooth representation of $H$.

Define the space $X$ of functions $f : G \to W$ which satisfy:

1. equivariant: $f(hg) = \sigma(h)f(g)$, $h \in H$, $g \in G$
2. smoothness: there is a compact open subgroup $K$ of $G$ (depending on $f$) such that $f(gx) = f(g)$, for $g \in G$, $x \in K$ (so $f$ is smooth i.e. continuous with $W$ having the discrete topology)

Then define a homomorphism $\Sigma : G \to \mathrm{Aut}_{\mathbb{C}}(X)$ by

$$\Sigma(g)f : x \longmapsto f(xg), \quad g, x \in G.$$

The pair $(\Sigma, X)$ provides a smooth representation of $G$ and is said to be *smoothly induced by* $\sigma$. It is usually denoted

$$(\Sigma, X) = \mathrm{Ind}_H^G \sigma.$$

**Proposition 1.2.20: Induced Representation is A Functor**

The map $\sigma \mapsto \mathrm{Ind}_H^G \sigma$ gives a functor $\mathrm{Rep}(H) \to \mathrm{Rep}(G)$.

***Proof*:**
exercise

There is a canonical $H$-homomorphism

$$\alpha_\sigma : \mathrm{Ind}_H^G \sigma \longrightarrow W, \quad f \longmapsto f(1).$$

The pair $(\mathrm{Ind}_H^G, \alpha)$ has the following fundamental property:

**Theorem 1.2.21: Frobenius Reciprocity**

Let $H$ be a closed subgroup of a locally profinite group $G$. For a smooth representation $(\sigma, W)$ of $H$ and a smooth representation $(\pi, V)$ of $G$, the canonical map

$$\begin{gathered} \mathrm{Hom}_G(\pi, \mathrm{Ind}_H^G \sigma) \longrightarrow \mathrm{Hom}_H(\pi, \sigma) \\ \varphi \mapsto \alpha_\sigma \circ \varphi, \end{gathered}$$

is an isomorphism that is functorial in both variables $\pi$, $\sigma$.

***Proof*:**
Let $f : V \to W$ be an $H$-homomorphism. We define a $G$-homomorphism $f_* : V \to \mathrm{Ind}\sigma$ by letting $f_*(v)$ be the function $g \mapsto f(\pi(g)v)$. The map $f \mapsto f_*$ is then the inverse of the above map.

The following shows how this is a very useful functor:

**Proposition 1.2.22: Induced Representation is Additive and Exact**

The functor $\mathrm{Ind}_H^G : \mathrm{Rep}(H) \to \mathrm{Rep}(G)$ is additive and exact.

need to re-phrase this!

***Proof*:**
For a smooth representation $(\sigma, W)$ of $H$, temporarily let $I(\sigma)$ denote the space of functions $G \to W$ satisfying the first condition $f(hg) = \sigma(h)f(g)$ of the definition above. Thus $I$ is a functor to the category of abstract representations of $G$; it is clearly additive and exact, while $\mathrm{Ind}_H^G(\sigma) = I(\sigma)^\infty$. Thus $\mathrm{Ind}_H^G$ is surely additive, and 2.3 Exercise (3) shows it to be left-exact.

To prove it is right-exact, let $(\sigma, W)$, $(\tau, U)$ be smooth representations of $H$ and let $f : W \to U$ be an $H$-surjection. Take $\varphi \in I(\tau)^\infty$, and choose a compact open subgroup $K$ of $G$ which fixes $\varphi$. The support of $\varphi$ is a union of cosets $HgK$, and the value $\varphi(g) \in U$ must be fixed by $\tau(H \cap gKg^{-1})$. By 2.3 Corollary 1 (applied to the group $H$ and the trivial representation of its compact open subgroup $H \cap gKg^{-1}$), there exists $w_g \in W$, fixed by $\sigma(H \cap gKg^{-1})$, such that $f(w_g) = \varphi(g)$. We define a function $\Phi : G \to W$ to have the same support as $\varphi$ and $\Phi(hgk) = \sigma(h)w_g$, for each $g \in H\backslash \mathrm{supp}\varphi/K$. The function $\Phi$ is fixed by $K$, and hence lies in $I(\sigma)^\infty$. Its image in $I(\tau)^\infty$ is $\varphi$, as required.

### Example 1.2.23: Induced Representation

1. here
2. here

As the compact open subgroups are the key subgroups to consider for this book, there is the following useful variation of induced representation.

### Definition 1.2.24: Compact Induction

Given $(\sigma, W)$ for a representation of closed $H \subseteq G$ and similarly defined $X$ as in definition 1.2.19, define:

$$X_c = \{f \in X \ : \ \text{compactly supported modulo } H\}$$

that is the image of $\text{supp}(f)$ in $H \setminus G$ is compact (equivalently, $\text{supp}(f) \subseteq HC$ for some compact set $C \subseteq G$). The induced functor is denoted $\text{c-Ind}_H^G \sigma$.

### Proposition 1.2.25: Compact Induction is Well-defined

The space $X_c$ along with $\text{c-Ind}_H^G \sigma$ is a smooth represetnation

***Proof*:**
exercise.

### Proposition 1.2.26: Compact Induction Repreis Additive and Exact

The functor $\text{c-Ind}_H^G : \text{Rep}(H) \to \text{Rep}(G)$ is additive and exact.

***Proof*:**
exercise.

### Example 1.2.27: Compact Induction

1. here
2. here

There is a natural $G$-embedding:

$$\text{c-Ind}_H^G(\sigma) \to \text{Ind}_H^G(\sigma) \qquad \text{(i.e. a natural transformation c-Ind}_H^G \Rightarrow \text{Ind}_H^G)$$

This map is an isomorphism if and only if $H \setminus G$ is compact. [2] says that there are examples of such a map being an isomorphism even when $H \setminus G$ is not compact with the right choice of $G, H, \sigma$.

The map is mostly interesting when $H \subseteq G$ is open in which case there is a natural $H$-homomorphism:

$$\alpha_\sigma^c : W \to \text{c-Ind}(\sigma) \qquad w \mapsto f_w$$

go over the Frobenius for compact; skipping for now because I have too little intuition

**1.2.28 Hypothesis From Now On** For any compact open $K \subseteq G$, $G/K$ is countable.

If $G/K$ is countable, so is $G/K'$ for any other compact open; for $K \cap K'$ is compact open and a finite index of $K$ thus

$$\frac{G}{K \cap K'} \to G/K$$

has finite fibers, thus $G/K \cap K'$ and $G/K'$ are both countable. All groups that will be considered in this book are have this property, but a few further results bellow don't require them. The main point is so that:

**Lemma 1.2.29: Irreducible Smooth Representation At Most Countable**

Let $(\pi, V)$ be a smooth irreducible representation of a locally profinite group $G$. Then $\dim_{\mathbb{C}}(V)$ is countable

***Proof*:**
Let $0 \neq v \in V$ and choose $K \subseteq G$ such that $v \in V^K$. As $V$ is irreducible,

$$\{\pi(g) \cdot v \ : \ g \in G/K\}$$

spans $V$, and this set is countable completing the proof.

This also allows for the following:

**Theorem 1.2.30: Schur's Lemma For Smooth Representations**

Let $(\pi, V)$ be a smooth irreducible representation of a locally profinite group $G$. Then

$$\mathrm{End}_{\mathbb{C}}(V) = \mathbb{C}$$

***Proof*:**
Let $0 \neq \varphi \in \mathrm{End}_{\mathbb{C}}(V)$. As the image and kernel of $\varphi$ must be $G$-subspaces, by irreducibility it must be bijective (which is enough in this category for invertible). Thus, $\mathrm{End}_{\mathbb{C}}(V)$ is at least a division $\mathbb{C}$-algebra.

Next, $\mathrm{End}_{\mathbb{C}}(V)$ must have countable dimension for if $0 \neq v \in V$, the $G$-translate of $v$ spans $V$, and so $\varphi \in \mathrm{End}_{\mathbb{C}}(V)$ is uniquely determined by the value of $\varphi(v)$. Now, if $\varphi \in \mathrm{End}_{\mathbb{C}}(V)$ where $\varphi \tilde{\in} \mathbb{C}$, $\varphi$ would be transcendental over $\mathbb{C}$ and hence the subset

$$\left\{(\varphi - a)^{-1} \ : \ a \in \mathbb{C}\right\}$$

would be linearly independent, and hence $\mathrm{End}_{\mathbb{C}}(V)$ would have uncountable $\mathbb{C}$-dimension, a contradiction. Thus, $\mathrm{End}_{\mathbb{C}}(V) \cong \mathbb{C}$, as we sought to show.

Note that this is an if and only if when working with finite groups. If $G$ is compact, this becomes an if and only if. This is false in general ([2] cites section 9.10).

**Corollary 1.2.31: Central Character**

Let $(\pi, V)$ be an irreducible smooth representation of $G$ and $Z = Z(G)$ is the center of $G$. Then $Z$ acts on $V$ via the character

$$\omega_\pi : Z \to \mathbb{C}^* \qquad \text{i.e.} \qquad \pi(z) \cdot v = \omega_\pi(z) \cdot v$$

for all $v \in V$ and $z \in Z$

***Proof*:**
As $z \in G$, the left-multiplication map is an endomorphism, hence $\pi(z) \in \mathrm{End}_G(V) \cong G$. Thus $\pi(z)$ is some scalar multiple of $\mathrm{id}_V$, hence $\pi(z) = \omega_\pi(z)$. To shows it's a character, note that if $K$ is a compact open subgroup of $G$ such that $V^K \neq 0$, Then $\omega_\pi$ is trivial om the compact open subgroup $K \cap Z$, showing it is indeed a smooth representation, and hence $\omega_\pi$ is a character of $Z$.

**Definition 1.2.32: Central Character**

The function $Z \to \mathbb{C}^*$ defined in corollary 1.2.31 is called a *central character*.

**Definition 1.2.33: Smooth Representation Admitting Central Character**

Let $(\pi, V)$ be a smooth representation of $G$ and $\chi$ a central character of $Z(G)$. Then we say that $(\pi, V)$ *admits* $\chi$ *as a central character* if:

$$\pi(z) \cdot v = \chi(z) \cdot v \qquad \forall v \in V,\ z \in V$$

**Proposition 1.2.34: Irreducible Smooth Representation of Abelian Groups**

Let $G$ be an abelian locally profinite group. Then an smooth irreducible representation is one-dimensional

***Proof*:**
By corollary 1.2.31, any representation $\pi$ will act like the character at every element $g \in G$ since $Z(G) = G$. Now, if $W$ is any one-dimensional subspace, as $\pi$ acts by scaling, $W$ is a $G$-subspace. Hence, any irreducible subspace must be one-dimensional, completing the proof.

**Proposition 1.2.35: KZ Space**

Let $(\pi, V)$ be a smooth representation of $G$, admitting $\chi$ as a central character. Let $K$ be an open subgroup of $G$ such that $KZ/Z$ is compact. Then:

1. Let $v \in V$. The KZ-space spanned by $v$ is of finite dimension, and is a sum of irreducible KZ-spaces.

2. As representation of $KZ$, the space $V$ is semisimple.

***Proof*:**
The vector $v$ is fixed by a compact open subgroup $K_0$ of $K$. The set $KZ/K_0Z$ is finite, so the space $W$ spanned by $\pi(KZ)v$ has finite dimension. Then $W$ is $K_0Z$-semisimple. Since $v$ was chosen arbitrarily, we get 2 as we sought to show.

The final elementary concept to adders is the dual space and how a representation $(\pi, V)$ relates to it:

**Definition 1.2.36: Smooth Dual (Contragradient)**

here

this lemma is given next, may move it somewhere better. It is useful for the following discussion about smooth duals.

**Lemma 1.2.37: Semisimple With Finite-index and Induction**

Let $G$ be a locally profinite group, and let $H$ be an open subgroup of $G$ of finite index.

1. If $(\pi, V)$ is a smooth representation of $G$, then $V$ is $G$-semisimple if and only if it is $H$-semisimple.
2. Let $(\sigma, W)$ be a semisimple smooth representation of $H$. The induced representation $\mathrm{Ind}_H^G \sigma$ is $G$-semisimple.

***Proof*:**
Suppose that $V$ is $H$-semisimple, and let $U$ be a $G$-subspace of $V$. By hypothesis, there is an $H$-subspace $W$ of $V$ such that $V = U \oplus W$. Let $f : V \to U$ be the $H$-projection with kernel $W$. Consider the map

$$f^G : v \mapsto (G : H)^{-1} \sum_{g \in G/H} \pi(g) f(\pi(g)^{-1} v), \quad v \in V.$$

The definition is independent of the choice of coset representatives and it follows that $f^G$ is a $G$-projection $V \to U$. We then have $V = U \oplus \mathrm{Ker}\, f^G$ and $\mathrm{Ker}\, f^G$ is a $G$-subspace of $V$. Thus $V$ is $G$-semisimple (cf. 2.2 Proposition).

Conversely, suppose that $V$ is $G$-semisimple. Thus $G$ is a direct sum of irreducible $G$-subspaces (2.2), and it is enough to treat the case where $V$ is irreducible over $G$. As representation of $H$, the space $V$ is finitely generated and so admits an irreducible $H$-quotient $U$. Suppose for the moment that $H$ is a normal subgroup of $G$. By Frobenius Reciprocity (2.4.2), the $H$-map $V \to U$ gives a non-trivial, hence injective, $G$-map $V \to \mathrm{Ind}_H^G U$. As representation of $H$, the induced representation $\mathrm{Ind}_H^G U = \oplus_{g \in G/H} gU$ is a direct sum of $G$-conjugates of $U$ (cf. 2.5 Lemma). These are all irreducible over $H$, so $\mathrm{Ind}\, U$ is $H$-semisimple. Proposition 2.2 then implies that $V \subset \mathrm{Ind}\, U$ is $H$-semisimple.

In general, we set $H_0 = \bigcap_{g \in G/H} gHg^{-1}$. This is an open normal subgroup of $G$ of finite index. We have just shown that the $G$-space $V$ is $H_0$-semisimple; the first part of the proof shows it is $H$-semisimple.

This completes the proof of (1), and (2) follows readily from the same arguments.

## 1.3 Measures and Duality

The space $\mathbb{C}[G]$ is $\mathbb{C}$-algebra isomorphic to $(\mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C}), +, *)$ where $*$ is convolution. This requires summing over each $g \in G$. In this book, $G$ is a.e. infinite, and so we replace summation with an appropriate measure on $G$.

Let $G$ be a locally profinite group, $C_c^\infty(G)$ the space of functions $f : G \to \mathbb{C}$ which are locally constant and of compact support (i.e. smooth and of compact support). Let $f \in C_c^\infty(G)$. Then compact support and smoothness (locally constant), here exists open subgroup $K_1, K_2 \subseteq G$ such that

$$f(k_1 g) = f(g) = f(g k_2) \qquad \forall g \in G,\ k_i \in K_i$$

By letting $K = K_1 \cap K_2$, $f$ must be a finite linear combination of characteristic functions of double cosets $KgK$. Now, $G \curvearrowright C_c^\infty(G)$ by left translation and right translation:

$$\begin{cases} \lambda : g \mapsto (\lambda_g f : x \mapsto f(g^{-1}x)) \\ \rho : g \mapsto (\rho_g f : x \mapsto f(xg) \end{cases} \qquad x, g \in G,\ f \in C_c^\infty(G)$$

Note that the left action is the one investing since the action is on functions. To see this, apply the operator $\lambda_{g_2}$ to the function $f$. This produces a new function:

$$h = \lambda_{g_2} f$$

The definition of this new function $h$ is:

$$h(y) = (\lambda_{g_2} f)(y) = f(g_2 y)$$

Now we apply the operator $\lambda_{g_1}$ to $h$; call the new function $k$:

$$k = \lambda_{g_1} h = \lambda_{g_1}(\lambda_{g_2} f) \qquad \text{so} \qquad k(z) = (\lambda_{g_1} h)(z) = h(g_1 z)$$

therefore:

$$k(z) = f(g_2(g_1 z)) = f(g_2 g_1 z)$$

showing why the left input needs to be the one inverted. The reader should check that $(C_c^\infty(G), \lambda)$ and $(C_c^\infty(G), \rho)$ are both smooth $G$-representation.

**Definition 1.3.1: Haar Integral**

A *right Haar integral* on $G$ is a non-zero linear functional:

$$I : C_c^\infty(G) \to \mathbb{C}$$

such that

1. $I(\rho_g f) = I(f)$ for all $g \in G,\ f \in C_c^\infty(G)$
2. For all $f \in C_c^\infty(G)$ where $f \geq 0$, $I(f) \geq 0$.

And similarly for the left Haar integral.

**Proposition 1.3.2: Existence of Haar Integral**

There exists right Haar integral $I : C_c^\infty(G) \to \mathbb{C}$. A linear functional $I' : C_c^\infty(G) \to \mathbb{C}$ is a right Haar integral if and only if $I' = cI$ for some $c > 0$.

The same then applies to the left Haar integral.

some parts of this proof reference results I did not write down yet; I'll put the label from the textbook for now.

***Proof*:**
Let $K$ be a compact open subgroup of $G$. Denote by ${}^K C_c^\infty(G)$ the space of functions in $C_c^\infty(G)$ fixed by $\lambda(K)$. Then viewing ${}^K C_c^\infty(G)$ as $G$-space via right translation, it is then identical to the representation of $G$ compactly induced from the trivial representation $1_K$ of $K$:

$${}^K C_c^\infty(G) = c\text{-}\mathrm{Ind}_K^G 1_K.$$

Now, viewing $\mathbb{C}$ as the trivial $G$-space (and as a consequence $K$-space), by compact Frobenius reciprocity:

$$\dim_{\mathbb{C}} \mathrm{Hom}_G({}^K C_c^\infty(G), \mathbb{C}) = \dim_{\mathbb{C}} \mathrm{Hom}_K(\mathbb{C}, \mathbb{C}) = 1.$$

Next, let's show there exists a non-zero element $I_K \in \mathrm{Hom}_G({}^K C_c^\infty(G), \mathbb{C})$ such that $I_K(f) \geq 0$ whenever $f \geq 0$, and if $h_K$ is the characteristic function of $K$, then $I_K(h_K) > 0$. Indeed, for $g \in G$, let $f_g$ denote the characteristic function of $Kg$. The set of functions $f_g$, $g \in K\backslash G$, then forms a $\mathbb{C}$-basis of the space ${}^K C_c^\infty(G)$ (2.5 Lemma). The functional $I_K : f_g \mapsto 1$ has the required properties, noting that $h_K = f_1$.

Now, choose a descending sequence $\{K_n\}_{n\geq 1}$ of compact open subgroups $K_n$ of $G$ such that $\bigcap_n K_n = 1$. Then:

$$C_c^\infty(G) = \bigcup_{n\geq 1} {}^{K_n} C_c^\infty(G).$$

For each $n \geq 1$, there is a unique right $G$-invariant functional $I_n$ on ${}^{K_n} C_c^\infty(G)$ which maps the characteristic function of $K_n$ to $(K_1 : K_n)^{-1}$. We have $I_{n+1}|_{{}^{K_n} C_c^\infty(G)} = I_n$, and so the family $\{I_n\}$ gives a functional on $C_c^\infty(G)$ of the required kind. The uniqueness statement is immediate, completing the proof.

In the above, think $K = \mathbb{Z}_p$ so that $f_g \to 1$ so $\mu(\mathbb{Z}_p) = 1$. The rest of the compact sets are given by the translation invariance and taking the index $[H : K]$ when $K \subseteq H$. From this we get $\mu(a + p^n\mathbb{Z}_p) = \mu(p^n\mathbb{Z}_p) = p^{-n}$. This extends to define a left Haar integral on $\mathbb{Q}_p$. If we take a finite extension $F/\mathbb{Q}_p$, simply replace $p$ with $\mathfrak{p}$ in the above. If we take $\mathrm{GL}_k(\mathbb{Q}_p)$, recall the chain of compact open subsets:

$$K_m = \{g \in K_0 | g \equiv I \pmod{p^m}\} = I + p^m M_k(\mathbb{Z}_p)$$

where $K_0 = \mathrm{GL}_k(\mathbb{Z}_p)$ has measure 1, and you do the same process.

**Corollary 1.3.3: Left-Right Haar Integral**

For $f \in C_c^\infty(G)$, define $\check{f} \in C_c^\infty(G)$ by $\check{f}(g) = f(g^{-1})$, $g \in G$. The functional

$$I' : C_c^\infty(G) \longrightarrow \mathbb{C},$$

$$I'(f) = I(\check{f}),$$

is a left Haar integral on $G$. Moreover, any left Haar integral on $G$ is of the form $cI'$, for some $c > 0$.

***Proof*:**
immediate.

From this, we can define a measure called the *Haar measure*. Let $I$ be a left Haar integral on $G$ and $\emptyset \neq S \subseteq G$ a compact open subset of a locally profinite group.

$$\mu_G(S) = I(\Gamma_S)$$

where $\Gamma_S$ is the characteristic function of $S$. Then $\mu_G(S) > 0$ and $\mu_G$ is left-translation invariant:

$$\mu_G(gS) = \mu_G(S) \qquad \forall g \in G$$

The reader should check that this is indeed a measure (see [4] for a review of measure theory)

**Definition 1.3.4: Haar Measure**

The measure $\mu_G$ defined above is called the *Haar measure*.

The function $I$ is often written as:

$$I(f) = \int_G f(g) d\mu_G(g) \qquad f \in C_c^\infty(G)$$

with the usual abbreviation of:

$$\int_S f(x) d\mu_G(x) := \int_G \Gamma_S(x) f(x) d\mu_G(x)$$

**Definition 1.3.5: Unimodular Group**

Let $G$ be a locally profinite group (more generally a topological group with a Haar measure defined on it). Then $G$ is *unimodular* if and only if the left Haar integral and is a right Haar integral.

**Example 1.3.6: Left and Right Haar Measures**

1. put the standard ones and interesting ones here, including the ones alluded earlier.

This is well-defined when working with functions with compact support. However, there shall be times when we have non-compact induction $\mathrm{Ind}_H^G\sigma$ where $H\backslash G$ is not compact (in this case it is often countable). To that end, we extend the domain of integration of the Haar measure. Consider $f : G \to \mathbb{C}$ that is invariant under left translation by a compact subgroup $K \subseteq G$, and let $\mu_G$ be a left Haar measure on $G$. Then if the series:

$$\sum_{g\in K\backslash G} \int_{Kg} |f(x)| d\mu_G(x)$$

converges, so does the non absolute values version of the series, and hence we can define:

$$\int_G f(x)\mu_G(x) = \sum_{g\in K\backslash G} \int_{Kg} f(x) d\mu_G(x)$$

Notice we are taking right cosets because the integral is left-invariant, and hence would have been a sum of constants and hence necessarily diverge. Furthermore, as $\mu_G$ is translation invariant this series is independent of choice of open compact group as we can take $K = K_1 \cap K_2$ which has finite index with both $K_1$ and $K_2$.

Next, we shall often construct characters using tensor products necessitating extending a Haar measure. Let $G_1, G_2$ be locally profinite groups, and set $G = G_1 \times G_2$. Then $G$ is locally profinite. An element $\sum_{1\le i\le r} f_i^1 \otimes f_i^2$ of the tensor product $C_c^\infty(G_1) \otimes C_c^\infty(G_2)$ gives a function on $G$ by

$$\Phi(g_1, g_2) = \sum_i f_i^1(g_1) f_i^2(g_2).$$

Then $\Phi \in C_c^\infty(G)$. Importantly, this gives an isomorphism

$$C_c^\infty(G_1) \otimes C_c^\infty(G_2) \cong C_c^\infty(G)$$

Let $\mu_j$ be a left Haar measure on $G_j$, $j = 1, 2$. Then there is a unique left Haar measure $\mu_G$ on $G$ such that

$$\int_G f_1 \otimes f_2(g)\, d\mu_G(g) = \int_{G_1} f_1(g_1)\, d\mu_1(g_1) \int_{G_2} f_2(g_2)\, d\mu_2(g_2),$$

for $f_i \in C_c^\infty(G_i)$. One writes $\mu_G = \mu_1 \otimes \mu_2$.

For a general $f \in C_c^\infty(G)$, the function

$$f_1(g_1) = \int_{G_2} f(g_1, g_2)\, d\mu_2(g_2)$$

lies in $C_c^\infty(G_1)$. We have

$$\int_G f(g)\, d\mu_G(g) = \int_{G_1} f_1(g_1)\, d\mu_1(g_1),$$

and symmetrically: if $f$ is of the form $f_1 \otimes f_2$, this is obvious, and such functions span $C_c^\infty(G)$.

Next, we naturally work over irreducible representations and hence smooth functions of the form $f : G \to V$ that need to be integrated. Let $V$ be a complex vector space, and consider the space $C_c^\infty(G; V)$ of locally constant, compactly supported functions $f : G \to V$. This space is isomorphic to $C_c^\infty(G) \otimes V$ in the obvious way: a tensor $\sum_i f_i \otimes v_i$ gives the function

$$g \mapsto \sum_i f_i(g) v_i.$$

If $\mu_G$ is a left Haar measure on $G$, there is a unique linear map $I_V : C_c^\infty(G;V) \to V$ such that

$$I_V(f \otimes v) = \int_G f(g) d\mu_G(g) \cdot v.$$

and so define:

$$I_V(\varphi) := \int_G \varphi(g) d\mu_G(g), \quad \varphi \in C_c^\infty(G;V).$$

This has the same invariance properties as the Haar integral on scalar-valued functions.

Next, we shall have cases where the left and right Haar measure don't agree (recall example 1.3.6). By the uniqueness of the left/right Haar measure up to a constant, we can cleverly equate the left/right Haar integration up to a constant. Let $\mu_G$ be a left Haar measure on $G$. For $g \in G$, consider the functional

$$\begin{aligned} I_g : C_c^\infty(G) &\to \mathbb{C}, \\ f &\mapsto \int_G f(xg)\, d\mu_G(x). \end{aligned}$$

Then $I_g$ is a left Haar integral on $G$, so there is a unique $\delta_G(g) \in \mathbb{R}_+^\times$ such that

$$\delta_G(g) \int_G f(xg)\, d\mu_G(x) = \int_G f(x)\, d\mu_G(x),$$

for all $f \in C_c^\infty(G)$. The function $\delta_G$ is a homomorphism $G \to \mathbb{R}_+^\times$. It is trivial if $G$ is abelian; more generally, $\delta_G$ is trivial if and only if $G$ is unimodular. If $f$ is the characteristic function of a compact open subgroup $K$ of $G$, then $\delta_G$ is trivial on $K$ and hence a character of $G$.

**Definition 1.3.7: Haar Module of G**

The function $\delta_G$ as define above is the *module* of $G$.

Note that if G is compact, then $\delta_G = 1$ and $G$ is unimodular. In the general case, any character $G \to \mathbb{R}_+^*$ is trivial on compact subgroups, since $\mathbb{R}_+^*$ has only the trivial compact subgroup.

Using the module, the functional

$$f \mapsto \int_G \delta_G(x)^{-1} f(x)\, d\mu_G(x), \quad f \in C_c^\infty(G),$$

is a right Haar integral on $G$. The mnemonic

$$d\mu_G(xg) = \delta_G(g) d\mu_G(x)$$

would help on how the left/right measures "commute".

Next, many important irreducible representations shall be $G$-subspaces of simpler induced representations, very often by taking a character on a closed subgroup and inducing it. Let $H$ be a closed subgroup of $G$, with module $\delta_H$. Let $\theta : H \to \mathbb{C}^\times$ be a character of $H$. Consider the space of functions $f : G \to \mathbb{C}$ which are $G$-smooth under right translation, compactly supported modulo $H$, and satisfy the "induction" homogeneity property:

$$f(hg) = \theta(h) f(g), \quad h \in H, g \in G.$$

**Definition 1.3.8: Induction Space of Smooth Functions**

Label this space

$$C_c^\infty(H\backslash G, \theta)$$

and view it as a smooth $G$-space via right translation $\rho$.

Note that $C_c^\infty(H\backslash G, \theta) = \text{c-Ind}_H^G \theta$.

**Theorem 1.3.9: Compact Induction on Haar Measure**

Let $\theta : H \to \mathbb{C}^\times$ be a character of $H$. The following are equivalent:

1. There exists a non-zero linear functional $I_\theta : C_c^\infty(H\backslash G, \theta) \to \mathbb{C}$ such that $I_\theta(\rho_g f) = I_\theta(f)$, for all $g \in G$.
2. $\theta\delta_H = \delta_G|_H$.

When these conditions hold, the functional $I_\theta$ is uniquely determined up to constant factor.

***Proof*:**

(1) $\Rightarrow$ (2) Let $\mu_G, \mu_H$ be left Haar measures on $G, H$ respectively. Define a $G$-map

$$C_c^\infty(G) \to C_c^\infty(H\backslash G, \theta) \qquad f \mapsto \tilde{f} \qquad \text{where} \qquad \tilde{f}(g) = \int_H \theta\delta_H(h)^{-1} f(hg)\, d\mu_H(h).$$

This map satisfies

$$\widetilde{\lambda_k f} = \delta_H\theta(k)^{-1}\tilde{f},$$

for $k \in H$ and $f \in C_c^\infty(G)$. If this map is surjective, we shall be able to factor $I_\omega$ through this map, allowing to link the two moduli.

To that end, If $K$ is a compact open subgroup of $G$, the space $C_c^\infty(G)^K$ by a simple generalization of proposition 2.1.6 this space is spanned by the characteristic functions of cosets $gK, g \in G/K$. On the other hand, each coset $HgK$ supports, at most, a one-dimensional space of functions in $C_c^\infty(H\backslash G, \theta)^K$. These subspaces span $C_c^\infty(H\backslash G, \theta)^K$, so the map $f \mapsto \tilde{f}$ is surjective on $K$-fixed functions, proving surjectivity.

Suppose that the space $C_c^\infty(H\backslash G, \theta)$ admits a functional $I_\theta$ of the required kind. The map $f \mapsto I_\theta(\tilde{f})$ is then a non-trivial $G$-homomorphism

$$(C_c^\infty(G), \rho) \to \mathbb{C}.$$

However, the space $\text{Hom}_G(C_c^\infty(G), \mathbb{C})$ has dimension 1 by Frobenius Reciprocity:

$$\dim_{\mathbb{C}} \text{Hom}_G({}^K C_c^\infty(G), \mathbb{C}) = \dim_{\mathbb{C}} \text{Hom}_K(\mathbb{C}, \mathbb{C}) = 1.$$

and is spanned by a right Haar integral. Therefore, the function $I_\theta$ exists if and only if the right Haar integral $C_c^\infty(G) \to \mathbb{C}$ factors through the quotient map $C_c^\infty(G) \to C_c^\infty(H\backslash G, \theta)$. When it does hold, $I_\theta$ is determined up to a constant factor.

Now, the kernel of the map $f \mapsto \tilde{f}$ contains all functions $\lambda_h f - \delta_H \theta(h)^{-1} f$, for $h \in H$ and $f \in C_c^\infty(G)$. Applying the right Haar integral on $G$ to such functions, we get

$$\int_G (\lambda_h f(g) - \delta_H \theta(h^{-1}) f(g)) \delta_G(g)^{-1}\, d\mu_G(g)$$
$$= \big(\delta_G(h^{-1}) - \delta_H \theta(h^{-1})\big) \int_G f(g) \delta_G(g)^{-1}\, d\mu_G(g).$$

This vanishes identically if and only if $\delta_G(h^{-1}) = \delta_H \theta(h^{-1})$, for all $h \in H$. Thus (1) $\Rightarrow$ (2).

(2) $\Rightarrow$ (1) Take a function $f \in C_c^\infty(G)$ such that $\tilde{f} = 0$. The function $f$ is fixed by a compact open subgroup $K$, and it is enough to treat the case where $\operatorname{supp} f \subset HgK$, for some $g \in K$. Thus $f$ is a finite linear combination of the characteristic function of cosets $h_i g K$, $h_i \in H$. The condition $\tilde{f} = 0$ then amounts to

$$\mu_H(H \cap gKg^{-1}) \sum_i \theta\delta_H(h_i)^{-1} f(h_i g) = 0,$$

since $\theta\delta_H$ is trivial on the compact subgroup $H \cap gKg^{-1}$ of $H$. On the other hand,

$$\int_G f(x)\delta_G(x)^{-1} d\mu_G(x) = \mu_G(K)\delta_G(g)^{-1} \sum_i \delta_G(h_i)^{-1} f(h_i g).$$

If (2) holds, this will vanish, giving a well-defined non-zero functional $I_\theta$, as we sought to show.

When the conditions of the proposition hold, the character $\theta$ takes only positive real values. Let $f \in C_c^\infty(G)$ satisfy $f(g) \geq 0$, for all $g \in G$; we then have $\tilde{f}(g) \geq 0$ for all $g$. Consequently:

**Corollary 1.3.10: Existence of Compact Induction on Haar Measure**

Suppose that the conditions of theorem 1.3.9 hold. There is then a non-zero linear functional $I_\theta$ on $C_c^\infty(H\backslash G, \theta)$ such that:

1. $I_\theta(\rho_g f) = I_\theta(f)$, for $f \in C_c^\infty(H\backslash G, \theta), g \in G$;
2. $I_\theta(f) \geq 0$, for $f \in C_c^\infty(H\backslash G, \theta), f \geq 0$.

These conditions determine $I_\theta$ uniquely, up to a positive constant factor, and we write:

$$I_\theta(f) = \int_{H\backslash G} f(g)\, d\mu_{H\backslash G}(g), \quad f \in C_c^\infty(H\backslash G, \theta),$$

**Definition 1.3.11: Semi-invariant Measure**

The measure $\mu_{H\backslash G}$ is called a positive *semi-invariant measure* on $H\backslash G$.

Since $\theta = \delta_H^{-1}\delta_G|H$ is uniquely determined, there is no real need to refer to it again, and so it shall be taken as implicit.

This allows for taking the dual of a compact induced smooth representation:

**Theorem 1.3.12: Duality Theorem**

Let $\dot{\mu}$ be a positive semi-invariant measure on $H\backslash G$. Let $(\sigma, W)$ be a smooth representation of $H$. Then there is a natural isomorphism:

$$(c\text{-Ind}_H^G \sigma)^\vee \cong \mathrm{Ind}_H^G \delta_{H\backslash G} \otimes \check{\sigma},$$

depending only on the choice of $\dot{\mu}$.

This proof is transcribed. I will put it in my own words asap!

***Proof*:**

We view $\delta_{H\backslash G} \otimes \check{\sigma}$ as acting on the same space $\check{W}$ as $\check{\sigma}$. Let $(\check{w}, w) \mapsto \langle \check{w}, w \rangle$ be the evaluation pairing $\check{W} \times W \to \mathbb{C}$. Let $\varphi \in$ c-Ind $\sigma$, $\Phi \in \mathrm{Ind}\, \delta_{H\backslash G} \otimes \check{\sigma}$, and consider the function

$$f : g \mapsto \langle \Phi(g), \varphi(g) \rangle, \quad g \in G.$$

This lies in $C_c^\infty(H\backslash G, \delta_{H\backslash G})$, so we have a $G$-invariant pairing

$$\mathrm{Ind}\, \delta_{H\backslash G} \otimes \check{\sigma} \times \text{c-Ind}\, \sigma \to \mathbb{C},$$
$$(\Phi, \varphi) \mapsto \int_{H\backslash G} \langle \Phi(g), \varphi(g) \rangle \, d\dot{\mu}(g).$$

This induces a $G$-homomorphism $\mathrm{Ind}\, \delta_{H\backslash G} \otimes \check{\sigma} \to (\text{c-Ind}\, \sigma)^\vee$. It is clearly natural in $\sigma$; we show it is an isomorphism.

To do this, we take a compact open subgroup $K$ of $G$ and describe the space $(\text{c-Ind}\, \sigma)^K$ of $K$-fixed functions in c-Ind $\sigma$. The support of such a function, $f$ say, is a finite union of cosets $HgK$. The value $f(g) \in W$ is then fixed by the compact open subgroup $H \cap gKg^{-1}$ of $H$. We choose a set $\mathcal{G}$ of representatives for the coset space $H\backslash G/K$. For each $g \in \mathcal{G}$, we choose a basis $\mathcal{W}_g$ of the space $W^{H\cap gKg^{-1}}$. For each $g \in \mathcal{G}$ and each $w \in \mathcal{W}_g$, there is a unique function $f_{g,w} \in (\text{c-Ind}\, \sigma)^K$, supported on $HgK$, such that $f_{g,w}(g) = w$. The set

$$\{f_{g,w} : g \in \mathcal{G}, w \in \mathcal{W}_g\}$$

is then a basis of $(\text{c-Ind}\, \sigma)^K$. That is, the space $(\text{c-Ind}\, \sigma)^K$ consists of the finite linear combinations of functions $f_{g,w}$, $g \in \mathcal{G}$, $w \in \mathcal{W}_g$.

Let $\mathcal{W}_g^*$ denote the basis of $\check{W}_g = \mathrm{Hom}_{\mathbb{C}}(W^{H\cap gKg^{-1}}, \mathbb{C})$ dual to $\mathcal{W}_g$. For $g \in \mathcal{G}$, $\check{w} \in \mathcal{W}_g^*$, we define $f_{g,\check{w}} \in (\mathrm{Ind}\, \delta_{H\backslash G} \otimes \check{\sigma})^K$ in the same way as before. We then have that the space $(\mathrm{Ind}\, \delta_{H\backslash G} \otimes \check{\sigma})^K$ consists of all functions $f$ such that, for any $g \in \mathcal{G}$, the restriction $f|_{HgK}$ is a finite linear combination of functions $f_{g,\check{w}}$, $\check{w} \in \mathcal{W}_g^*$.

For $g_1, g_2 \in \mathcal{G}$, $w \in \mathcal{W}_{g_1}$, $\check{w} \in \mathcal{W}_{g_2}^*$, we have

$$\langle f_{g_2,\check{w}}, f_{g_1,w} \rangle = \begin{cases} \dot{\mu}(Hg_1K)\langle \check{w}, w \rangle & \text{if } Hg_1K = Hg_2K, \\ 0 & \text{otherwise,} \end{cases}$$

for some $\dot{\mu}(Hg_1K) > 0$. The pairing thus identifies $(\mathrm{Ind}\, \delta_{H\backslash G} \otimes \check{\sigma})^K$ with the linear dual of $(\text{c-Ind}\, \sigma)^K$, and hence $\mathrm{Ind}\, \delta_{H\backslash G} \otimes \check{\sigma}$ with $(\text{c-Ind}\, \sigma)^\vee$, as required.

## 1.4 Hecke Algebra

Linear representations of finite groups are in bijective correspondence to the modules over the group algebra $\mathbb{C}[G]$. A similar conditions holds for smooth representations of locally profinite groups given the suitable generalization of $\mathbb{C}[G]$ to Hecke algebras.

Though not necessary, to avoid technicalities that don't add insight, throughout this entire section the locally profinite group $G$ will be assumed to be unimodular.

Fix a Haar measure $\mu$ on $G$. Then for any $f_1, f_2 \in C_c^\infty(G)$, define the convolution:

$$f_1 * f_2(g) = \int_G f_1(x) f_2(x^{-1}g) d\mu(x)$$

This gives a function $(x,) \mapsto f_1(x) f_2(x^{-1}g) \in C_c^\infty(G \times G)$ and so by ref:HERE $f_1 * f_2 \in C_c^\infty(G)$. This operation is associative: Here's the equation in an align* environment:

$$\begin{aligned} f_1 * (f_2 * f_3)(g) &= \int \int f_1(x) f_2(y) f_3(y^{-1}x^{-1}g) d\mu(y) d\mu(x) \\ &= \int \int f_1(x) f_2(x^{-1}y) f_3(y^{-1}g) d\mu(y) d\mu(x) \\ &= \int \int f_1(x) f_2(x^{-1}y) f_3(y^{-1}g) d\mu(x) d\mu(y) \\ &= (f_1 * f_2) * f_3(g) \end{aligned}$$

**Definition 1.4.1: Hecke Algebra**

Let $G$ be a unimodular locally profinite group. Then:

$$\mathcal{H}(G) = (C_c^\infty(G), *)$$

is is an associative $\mathbb{C}$-algebra called the *Hecke algebra* of $G$.

This is an example of a *non-unital* algebra if $G$ is not finite. If $G$ is commutative, so is $\mathcal{H}(G)$. Note that if $\mu, \nu$ are two choices of Haar measures giving rise to $\mathcal{H}_\mu(G)$ and $\mathcal{H}_\nu(G)$ on $C_c^\infty(G)$, then there exists a constant $c > 0$ such that $\nu = c\mu$ and a map $f \mapsto c^{-1} f$, and hence

$$\mathcal{H}_\mu(G) \cong \mathcal{H}_\nu(G)$$

The actual correspondence is proven here, going to write out the details later.

Goal: An $\mathcal{H}(G)$-module is smooth if $\mathcal{H}(G) * M = M$. If $M$ is a smooth $\mathcal{H}(G)$-module, then there exists a homomorphism $\pi : G \to \mathrm{Aut}_\mathbb{C}(M)$ such that $(\pi, M)$ is a smooth representation. A vice-versa.

## 1.5 Important Special Case: Representation of $\mathrm{GL}_2(\mathbb{F}_p)$

When analyzing $\mathrm{GL}_2(\mathbb{Q}_p)$ and related groups, we shall draw inspiration of this special case. We shall first single out the following subgroups:

**Definition 1.5.1: Matrix Subgroups**

Let $F$ be (an arbitrary) field. Then define the groups:

1. (Bruhat subgroup) $B = \left\{\begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \in \mathrm{GL}_2(F)\right\}$
2. (unipotent subgroup) $N = \left\{\begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} \in \mathrm{GL}_2(F)\right\}$
3. (torus subgroup) $T = \left\{\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \in \mathrm{GL}_2(F)\right\}$
4. (center) $Z = \left\{\begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix} \in \mathrm{GL}_2(F)\right\} \cong F^*$

**Lemma 1.5.2: Bruhat Decomposition**

Let $G = \mathrm{GL}_2(F)$. Then its *Bruhat decomposition* is:

$$G = B \cup BwB$$

where

$$w = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Hence, $\{1, w\}$ is a set of representative of the coset space $B\backslash G/B$.

Furthermore, $B = NT = TN$, hence $B \cong N \rtimes T$

***Proof*:**
Let $M \in GL_2(F)$ so that $\det(M) = ad - bc \neq 0$ if $c = 0$, $ad \neq 0$, so $a, d \neq 0$. Then $M \in B$.

If $c \neq 0$, choose two arbitrary $B_1, B_2 \in B$, calculate $B_1 w B_2$, use the fact that $\det(B_1), \det(B_2) \neq 0$ to get an equation for each entry of $M$. Note that the bottom-left entry *must* be nonzero and hence $B \cap BwB = \emptyset$, proving the decomposition into disjoint cosets.

For the $B = NT = TN$, notice that:

$$\begin{pmatrix} a & b \\ 0 & d \end{pmatrix} = \begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix} \begin{pmatrix} 1 & b/a \\ 0 & 1 \end{pmatrix} \in TN$$

It can also be factored as:

$$\begin{pmatrix} a & b \\ 0 & d \end{pmatrix} = \begin{pmatrix} 1 & b/d \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix} \in NT$$

Showing they are a semidirect products completing the proof.

By constructing, any representative of the coset $BwB$ has a nonzero entry in the $(2, 1)$-position. As

$B = TN = NT$ and $w$ normalizes $T$ ($wTw = T$):

$$BwB = NwB = BwN$$

Furthermore, the map:

$$B \times N \to BwN \qquad (b, n) \mapsto bwn$$

is *bijective.*

Next, For any $M \in \mathrm{GL}_2(F)$, define $\mathrm{ch}_M(t) = \det(tI - M) = t^2 - \mathrm{tr}(M)t + \det(M)$, that is $\mathrm{ch}_M(t)$ is the characteristic polynomial of $M$, i.e. a monic quadratic $F$-polynomial and $ch_M(0) = \det(M) \neq 0$.

**Proposition 1.5.3: Character Polynomial For $\mathrm{GL}_2(F)$**

Let $F$ be an arbitrary field, $M \in \mathrm{GL}_2(F)$, and $f(t) = \mathrm{ch}_M(t) = \det(tI - M)$. Then:

1. If $f(t)$ is irreducible, the $F$-algebra $F[M] \subseteq M_2(F)$ is a field with $\mathrm{GL}_2(F)$-centralizer $F[M]^*$. Given $f(t) = t^2 + at + b$, then $M$ is $\mathrm{GL}_2(F)$-conjugate to:

$$\begin{pmatrix} 0 & -b \\ 1 & -a \end{pmatrix}$$

   Furthermore, an element $N \in \mathrm{GL}_2(F)$ is $\mathrm{GL}_2$-conjugate go $M$ if and only if $\mathrm{ch}_N(t) = f(t)$.

2. If $f(t)$ has distinct roots $a, b \in F^*$, then $M$ is $\mathrm{GL}_2(F)$-conjugate to:

$$\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$$

3. If $f(t)$ has a repeated root $a \in F^*$, Then $M$ is $\mathrm{GL}_2(F)$-conjugate to either:

$$\begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix} \qquad \text{or} \qquad \begin{pmatrix} a & 1 \\ 0 & a \end{pmatrix}$$

   In the first case, $M$ lies in the center of $\mathrm{GL}_2(F)$, and in the 2nd the $\mathrm{GL}_2(F)$-centralizer of $M$ is $F[M]^*$.

***Proof*:**
We shall do one and leave the rest as an exercise.

1. As $f(t)$ is irreducible, by Cayley-Hamilton $F[M] \cong F[t]/(f(t))$ where the characteristic polynomial $f(t)$ is also the minimal polynomial. For the conjugacy result, as $M$ is irreducible it has no eigenvalues and so $\{v, Mv\}$ forms a basis for any $v \in F^2$. Finding the matrix representation of the linear transformation corresponding to $M$ shall give the desired result

2. exercise

3. exercise

Now let $F = \mathbb{F}_q$ be a field with $q = p^l$ elements for some $l \in \mathbb{N}$, and let $G = \mathrm{GL}_2(\mathbb{F}_q)$. Let us find the irreducible complex representations of $G$. First off:

**Lemma 1.5.4: Order and Irreducible Representations of $G$**

Let $G = \mathrm{GL}_2(\mathbb{F}_q)$. Then:

$$|G| = (q^2 - 1)(q^2 - q)$$

Furthermore, $G$ has $q^2 - 1$ conjugacy classes.

Thus, there are $q^2 - 1$ irreducible representations.

***Proof*:**
Let the first column be any nonzero vector, giving $q^2 - 1$ possibilities. The second column cannot be a scalar multiple of the first. Given a choice $v_1$ of the first column, the second cannot be $kv_1$ for all $k \in \mathbb{F}_q$. This gives $q^2 - q$ possibilities. These choices are linearly independent, thus:

$$|G| = (q^2 - 1)(q^2 - q)$$

For the conjugacy classes we repeatedly use proposition 1.5.3. By the first case, as $\mathrm{GL}_2(\mathbb{F}_q)$ has exactly $(q^2 - q)/2$ irreducible monic polynomials of degree 2, there are $(q^2 - q)/2$ conjugacy classes. The second case will give $(q - 1)(q - 2)/2$ cases, and the third $2(q - 1)$ cases. Summing these up gives the desires result.

Thus, there are $q^2 - 1$ irreducible representations. As $G$ is finite, we may take advantage of character theory, hence we shall construct some characters. First, note that $N \cong (\mathbb{F}_q, +)$ via $x \mapsto \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}$.

Fix a nontrivial character of $(\mathbb{F}_q, +)$ $\chi$ (ex. $\chi(x) = e^{2\pi i (x)/p}$). Then $(a\chi)(x) = \chi(ax)$ ranges over all characters of $(\mathbb{F}_q, +)$. From this, we can define a natural action of conjugation by $T$ (or $B$) on $N$. Namely the action to $T$ on $N$ produces an action on the characters:

$$(t\chi)(n) = \chi(t^{-1}nt)$$

when interpreting $n$ as a matrix in $N$ via the isomorphic, $N \cong (F_q, +)$ given above. Certainly all non-trivial characters are in the orbit of $T$, hence the conjugacy classes split into the single trivial character and all other ones.

**1.5.5 Non-trivial Characters of N** Therefore, all non-linear character characters on $N$ are similar up to conjugation.

The trivial one shall be important in a moment in finding our first irreducible representation.

We first define the following: let $\chi_1, \chi_2$ be characters of $\mathbb{F}_q^*$. Define the character:

$$\chi = \chi_1 \otimes \chi_2 : \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \mapsto \chi_1(a)\chi_2(b)$$

This is a character of $T$. We may regard this as a character on $B$ that is trivial on $N$ via the quotient:

$$B \to B/N \cong T$$

This process is given a name:

**Definition 1.5.6: Inflation**

Let $H \trianglelefteq G$ be a normal subgroup and $(\pi, V)$ a representation of $G/H$. Then the *inflation* of $\pi$ from $H$ to $G$ is a new $\tilde{\pi} : G \to V$ such that $\tilde{\pi}(gH) = \pi(\overline{g})$; that is, it is trivial on the cosets of $G/H$.

After inflating from $T$ to $B$, we may form the induced representation of $\mathrm{GL}_2(\mathbb{F}_q) = G$

$$\mathrm{Ind}_B^G \chi$$

and consider its irreducible components:

**Lemma 1.5.7: Irreducible Components of G**

Let $\pi$ be an irreducible representation of $G$. Then the following are equivalent:

($\Rightarrow$) $\pi$ is equivalent to a $G$-subspace of $\mathrm{Ind}_B^G \chi$ for a character $\chi$ of $T$ seen as a character on $B$ trivial on $N$.

($\Rightarrow$) $\pi$ contains the trivial character of $N$.

***Proof*:**
First, $\pi$ contains the trivial character of $N$ if and only if it is contains an irreducible representation $\sigma$ of $B$ containing to the trivial character of $N$. Indeed, restrict $G$ to $B$, $\mathrm{Res}_B^G \pi$ and then restrict further to $N$

$$\mathrm{Res}_N^B(\mathrm{Res}_B^G \pi)$$

By Maschke's theorem breakdown $\mathrm{Res}_B^G \pi$ into irreducibles. Call this irreducible representation of $B$ $\sigma$. As Res is additive over direr sums, we see the above if and only if holds. Now as we just proved above this is equivalent to $\sigma$ being an inflation of a character $T$. By Frobenius reciprocity:

$$\mathrm{Hom}_G(\mathrm{Ind}_B^G \chi, \pi) \cong \mathrm{Hom}_B(\chi, \mathrm{Res}_B^G \pi)$$

As $\pi$ is irreducible we can apply the same ideas as Schur's Lemma to finish the proof.

So, if $\pi$ contains a trivial character of $N$, it is irreducible! By the lemma, we analyze the inced representations $\mathrm{Ind}_B^G \chi$. To start, define the character of $T$:

$$\chi^w : \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \mapsto \chi_2(a)\chi_1(b)$$

where $w$ is the permutation in $G$ swapping columns. By abusing notation, we can say this is a character if $B$ that is trivial on $N$.

**Proposition 1.5.8: Analyzing Inflation of Torus Character**

Let $\chi, \xi$ be character of $T$ viewed as characters of $B$ trivial on $N$. Then:

1. The space $\mathrm{Hom}_G(\mathrm{Ind}_B^G(\chi), \mathrm{Ind}_B^G(\xi))$ is trivial, unless $\chi = \xi$ and $\chi = \xi^w$.
2. The space

$$\mathrm{Hom}_G(\mathrm{Ind}_B^G(\chi), \mathrm{Ind}_B^G(\chi)) \quad \text{and} \quad \mathrm{Hom}_G(\mathrm{Ind}_B^G(\chi), \mathrm{Ind}_B^G(\chi^w))$$

have the same dimension: it is 2 if $\chi = \chi^w$ and 1 otherwise.

***Proof*:**
We use the canonical isomorphism of Frobenius Reciprocity

$$\mathrm{Hom}_G(\mathrm{Ind}_B^G(\chi), \mathrm{Ind}_B^G(\xi)) \cong \mathrm{Hom}_B(\mathrm{Res}_B^G(\mathrm{Ind}_B^G(\chi)), \xi) \tag{1.1}$$

The restriction-induction formula of Mackey gives

$$\mathrm{Res}_B^G \mathrm{Ind}_B^G(\chi) \cong \sum_{y \in B \backslash G / B} \mathrm{Ind}_{B \cap B_y}^B \left( \mathrm{Res}^{B_y \cap B}(\chi^y) \right), \tag{1.2}$$

where $\chi^y$ denotes the character $b \mapsto \chi(yby^{-1})$ of the group $B^y = y^{-1}By$. By Bruhat decomposition we only have to consider the cases $y = 1$ and $y = w$. The term in equation (1.2) corresponding to $y = 1$ is just $\chi$, and so contributes a factor $\mathrm{Hom}_T(\chi, \xi)$ to equation (1.1). The term corresponding to $w$ contributes

$$\mathrm{Hom}_B(\mathrm{Ind}_T^B(\chi^w), \xi) \cong \mathrm{Hom}_T(\chi^w, \xi).$$

Altogether, equation (1.1) reduces to

$$\mathrm{Hom}_B(\mathrm{Ind}_B^G(\chi), \mathrm{Ind}_B^G(\xi)) \cong \mathrm{Hom}_T(\chi, \xi) \oplus \mathrm{Hom}_T(\chi^w, \xi),$$

and the result follow from Schur's lemma, completing the proof.

**Corollary 1.5.9: Irreducible Representations Given Torus Character**

Let $\chi$ be a character of $T$ viewed as a character of $B$ trivial on $N$. Then:

1. The representation $\mathrm{Ind}_B^G \chi$ is irreducible if and only if $\chi \neq \chi^w$
2. If $\chi = \chi^w$, then $\mathrm{Ind}_B^G \chi$ has length 2 with distinct component factors

***Proof*:**
Apply Schur's lemma to the above proposition.

**Example 1.5.10: Induced Character of Trivial Character**

Consider the trivial character $1_B$ of $B$. The trivial character $1_G$ occurs in $\mathrm{Ind}_B^G 1_B$, so

$$\mathrm{Ind}_B^G 1_B = 1_G \oplus \mathrm{St}_G,$$

for a unique irreducible representation $\mathrm{St}_G$, called the Steinberg representation. Clearly, $\dim \mathrm{St}_G = q$ since $[G : B] = q + 1$ and $\dim 1_G = 1$.

These irreducible representations are given a name:

**Definition 1.5.11: Principle Series Representations**

The representation $\mathrm{Ind}_B^G$ of a character $\chi$ of $T$ is called the *principal series*. The term principle series is often extended to include the irreducible components, namely all irreducible representations of $G$ containing the trivial character of $N$.

**Corollary 1.5.12: Counting Principal Series**

Up to isomorphism, the group $\mathrm{GL}_2(F)$ has precisely $\frac{1}{2}(q^2 + q) - 1$ irreducible representations which contain the trivial character of $N$.

***Proof*:**

This is a counting argument: if $\chi = \chi^w$, there are $2(q-1)$ such characters (2 for the length 2). If they are not equal there are $\frac{1}{2}\cdot[(q-1)^2-(q-1)]$ (where we half since $\mathrm{Ind}_B^G\chi_1\otimes\chi_2) = \mathrm{Ind}_B^G\chi_2\otimes\chi_1)$). Adding them gives the desired result.

Notice this number is less than $q^2 - 1$, meaning not all irreducible representations are of this form. In particular there are $\frac{1}{2}(q^2 - q)$ such irreducible representations. An irreducible representation need not be trivial on $N$ as we shall be given a name:

**Definition 1.5.13: Cuspidal Representation**

An irreducible representation of $\mathrm{GL}_2(F)$ not containing the trivial character of N is called a *cuspidal representation*.

A cuspidal representation $\sigma$ must contain some non-trivial character of $N$. As it is non-trivial on $N$, any two non-trivial characters of $N$ are $B$-conjugate (see remark 1.5.5), so $\sigma$ contains all non-trivial characters of $N$. This adds complexity, in particular the cuspidal representations cannot be constructed directly by induction.

To to that end, let $L/\mathbb{F}_q$ be a quadratic extension so that it has $q^2$ elements and $L^*$ has $q^2 - 1$ elements. Let $F = \mathbb{F}_q$ to simply notation. The characters $\theta$ of $L^*$ correspond to the $(q^2-1)$ primitive roots of unity, of which there are $q^2 - 1 - 1 = q^2 - 2$. A character is called regular if it is *not* invariant under the Frobenius automorphism, that is $\theta(x^q) \neq \theta(x)$. A character $\theta_j(g^k) = \omega^{jk}$ is *not regular* if $j$ divides $q + 1$, as can be quickly verified. There are $(q^2 - 1)/(q + 1) = q - 1$ non-regular characters, and hence $q(q - 1)$ regular characters. Choose a basis of $L$ so that $L \cong_F F$ and $\mathrm{GL}_2(F)$ is associated with $\mathrm{Aut}_F(L)$. There is a natural action $L^*$ on $L$, which corresponds to a subgroup of

$\mathrm{GL}_2(F) \cong \mathrm{Aut}_F(L)$. Identify:

$$L^* \stackrel{\simeq}{=} E \subseteq \mathrm{GL}_2(F)$$

Importantly, note that the elements of $G$ with irreducible characteristic polynomial is conjugate to an element of $E$. For, if $0 \neq \lambda \in L$, then the corresponding matrix in $\mathrm{GL}_2(F)$ must have $\lambda$ as its eigenvalue, and as it must be irreducible the other root must be the galois conjugate $\lambda^q$, but this shows the correspondence.

Now, let $\theta$ be a regular character of $E$ and $\psi$ a non-trivial character of $N$. Define $\theta_\psi$ of $ZN$ as:

$$\theta_\psi : \begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix} u \mapsto \theta(a)\psi(u)$$

Note that $\mathrm{Ind}_{ZN}^G \theta_\psi$ is independent of choice of $\psi$ up to equivalence (see remark 1.5.5).

**Theorem 1.5.14: Cuspidal Representation of Finite Matrix Group**

Let $\theta$ be a regular character of $E$ and $\psi$ a non-trivial character of $N$.

1. The virtual representation
$$\pi_\theta = \mathrm{Ind}_{ZN}^G \theta_\psi - \mathrm{Ind}_E^G \theta$$
is an irreducible cuspidal representation of $G$ of dimension $q-1$.
2. Let $\theta_1, \theta_2$ be regular characters of $E$; then $\pi_{\theta_1} \cong \pi_{\theta_2}$ if and only if $\theta_2 = \theta_1$ or $\theta_2 = \theta_1^q$.
3. Every irreducible cuspidal representation of $G$ is of the form $\pi_\theta$, for some regular character $\theta$ of $E$.

***Proof*:**

1. As we are working with finite groups, representations are characterized by characters. The character of the virtual representation $\pi_\theta$ is given by the difference of the characters of the two induced representations:
$$\mathrm{tr}\pi_\theta(g) = \mathrm{tr}(\mathrm{Ind}_{ZN}^G \theta\psi)(g) - \mathrm{tr}(\mathrm{Ind}_E^G \theta)(g)$$
We use the general formula for the character of an induced representation, namely if $(\sigma, W)$ is a representation of a subgroup $H \leq G$, then the character of $\Pi = \mathrm{Ind}_H^G \sigma$ is:
$$\mathrm{tr}(\Pi(g)) = \frac{1}{|H|} \sum_{x \in G \text{ s.t. } x^{-1}gx \in H} \sigma(x^{-1}gx)$$
We compute this for each part and for each type of conjugacy class in $G = \mathrm{GL}_2(k)$. First take $\Pi_1 = \mathrm{Ind}_{ZN}^G \theta\psi$ so $H = ZN = B$ with $|H| = (q-1)q$. The character is $\sigma_1 = \theta\psi$.
   - Let $g = z \in Z$. Since $z$ is central, $x^{-1}zx = z$ for all $x \in G$. The element $z$ is in $ZN$, so the condition is always met.
$$\mathrm{tr}(\Pi_1(z)) = \frac{1}{(q-1)q} \sum_{x \in G} \theta\psi(z) = \frac{|G|}{(q-1)q}\theta(z) = \frac{(q^2-1)(q^2-q)}{(q-1)q}\theta(z) = (q^2-1)\theta(z)$$

- Let $g = zu$ with $z \in Z, u \in N, u \neq 1$. An element with a repeated eigenvalue is only conjugate to other elements with repeated eigenvalues. The conjugates of $g$ that lie in $ZN$ are only those of the form $z'u'$. A more detailed analysis of the character sum shows that the sum over the unipotent parts averages to $-1$ when paired against a non-trivial character $\psi$. This yields:

$$\mathrm{tr}(\Pi_1(zu)) = -\theta(z)$$

- Let $g = y \in E \setminus Z$. The characteristic polynomial of $y$ is irreducible over $k$. Every element of $ZN$ has its eigenvalues in $k$, so its characteristic polynomial is reducible. Thus, no conjugate of $y$ can be in $ZN$. The sum in the character formula is empty.

$$\mathrm{tr}(\Pi_1(y)) = 0$$

Next, consider $\Pi_2 = \mathrm{Ind}_E^G\theta$. Here $H = E^\times$ is a subgroup of order $q^2 - 1$. The character is $\sigma_2 = \theta$.

- Let $g = z \in Z$. Since $z \in F^\times \subset E^\times$, the condition $x^{-1}zx \in E^\times$ is always met. Next, consider $\Pi_2 = \mathrm{Ind}_E^G\theta$ Here $H = E^\times$, a subgroup of order $q^2 - 1$. The character is $\sigma_2 = \theta$.
$$\mathrm{tr}(\Pi_2(z)) = \frac{1}{q^2-1}\sum_{x \in G}\theta(z) = \frac{|G|}{q^2-1}\theta(z) = (q^2 - q)\theta(z)$$

- Let $g = zu$ with $u \neq 1$. The element $g$ has a repeated eigenvalue. No conjugate of $g$ can have an irreducible characteristic polynomial, so no conjugate of $g$ lies in $E^\times \setminus Z$. Conjugation into $Z$ is impossible for $u \neq 1$. The sum is empty.

$$\mathrm{tr}(\Pi_2(zu)) = 0$$

- Let $g = y \in E \setminus Z$. The centralizer of $y$ in $G$ is $E^\times$. The conjugacy class of $y$ intersects $E^\times$ in exactly two elements: $y$ and its Galois conjugate $y^q$. The formula for the character of an induced representation on an element $h \in H$ simplifies in this case to $\sum_i \theta(g_i h g_i^{-1})$ where $g_i$ are representatives for $H\backslash G/H$. For $GL_2$, this gives:

$$\mathrm{tr}(\Pi_2(y)) = \theta(y) + \theta(y^q) = \theta(y) + \theta^q(y)$$

We now subtract the character values to get:

- For $g = z \in Z$:

$$\mathrm{tr}\pi_\theta(z) = (q^2 - 1)\theta(z) - (q^2 - q)\theta(z) = (q - 1)\theta(z)$$

- For $g = zu$, $u \neq 1$:
$$\mathrm{tr}\pi_\theta(zu) = -\theta(z) - 0 = -\theta(z)$$

- For $g = y \in E \setminus Z$:

$$\mathrm{tr}\pi_\theta(y) = 0 - (\theta(y) + \theta^q(y)) = -(\theta(y) + \theta^q(y))$$

This completes the derivation of the character:

| **Conjugacy Class Representative** $g$ | **Character Value** $\mathrm{tr}\pi_\theta(g)$ |
|---|---|
| $z \in Z$ | $(q-1)\theta(z)$ |
| $zu$ where $z \in Z, u \in N, u \neq 1$ | $-\theta(z)$ |
| $y \in E \setminus Z$ | $-(\theta(y) + \theta^q(y))$ |
| $g$ not conjugate to an element of $ZN \cup E$ | 0 |

Now, we can take the inner product of the character with itself to get:

$$\frac{1}{|G|}\sum_{g \in G} |\mathrm{tr}\pi_\theta(g)|^2 = 1$$

Now, recall that a representation $V$ is irreducible if given the basis of characters and the inner product with itself, we get 1. In this case, we may have gotten a virtual character, that is the coefficient was actually negative. This is easily checkable not to be the case by computing that

$$\mathrm{tr}\pi_\theta(e) = q - 1 = \dim(V)$$

If it were virtual, this would be negative, showing it is indeed a character and furthermore $\pi_\tau$ is an irreducible representation. Next, as

$$\sum_{u \in N} |\mathrm{tr}\pi_\theta(u)| = 0$$

we see that $\pi_\theta|_N$ does not contain the trivial character of $N$, and hence $\pi_\theta$ is cuspidal.

2. Two representations are isomorphic if they have the same character. It is easy to see from the character table above that $\pi_{\theta_1} \cong \pi_{\theta_2}$ if and only if $\theta_2 = \theta_1$ or $\theta_2 = \theta_1^q$.

3. As $E^*$ has $q^2 - q$ characters and the group $F$ has $q-1$ we get a total $q^2 - q - (q-1) = q^2 - q$ number of regular characters given of this form, which combining with corollary 1.5.12 gives us the total of $q^2 - 1$ irreducible representations, and hence we found all irreducible representations by lemma 1.5.4, completing the proof.

The book presents this interesting relation between characters and determinants as an exercise that I want to put down because it's pretty cool:

**Example 1.5.15: Character and Determinants**

Let $\psi$ be a non-trivial character of $N$, and consider the representation

$$F = \mathrm{Ind}_N^G \psi.$$

$\sigma$ be an irreducible representation of $G$. Show that:

1. If $\sigma = \varphi \circ \det$, for a character $\varphi$ of $k^\times$, then $\sigma$ does not occur in $F$.

2. Otherwise, $\sigma$ occurs in $F$ with multiplicity one.

# 2

# *Induced Representation (of Linear Groups)*

Now let $G = \mathrm{GL}_2(F)$ where $F$ is a nonarchemedian field, ex. $\mathbb{Q}_p$, $\mathbb{F}_q(t)$, or finite extensions of them. The goal is to classify all smooth irreducible representations of $G$.

## 2.1 Linear Groups over Local Fields

Given the additional structure, we have some more interesting decompositions. We have $B$, $N$, $T$, and $Z$ as before as these are define for any field $F$. Note that $B/N \cong T$ and $B \cong T \ltimes N$ are homeomorphisms.

**Proposition 2.1.1: Iwasawa Decompositon**

Let $B$ the be Borel subgroup and $K = \mathrm{GL}_2(\mathfrak{o})$. Then:

$$G = BK$$

***Proof*:**
Take $g \in G$; if the $(2,1)$-entry of $g$ is zero, then $g \in B$. Otherwise, post-multiplying by the permutation matrix $w \in K$ if necessary, we can assume $v_F(g_{21}) \geqslant v_F(g_{22})$. We can then post-multiply by a lower triangular matrix in $K$ to achieve $g_{21} = 0$.

**Corollary 2.1.2: Compact Quotient**

The quotient space

$$B \setminus G$$

is compact

***Proof*:**
$B \setminus G$ is the continuous image of the compact subgroup $K = \mathrm{GL}_2(\mathfrak{o})$.

**Proposition 2.1.3: Cartan Decomposition**

Let $\varpi$ be a prime element of $F$. Then the matrices:

$$\begin{pmatrix} \varpi^a & 0 \\ 0 & \varpi^b \end{pmatrix} \qquad a, b \in \mathbb{Z},\ a \leq b$$

form a set of representatives for the coset space $K \backslash G / K$.

***Proof*:**
Permuting rows and columns, using the permutation matrix in $K$, we can arrange for the largest entry of $g$ (in absolute value) to be in the $(1, 1)$-place. Multiplying by elementary matrices from $K$, we can then arrange for $g$ to be diagonal, and unit factors can be absorbed into $K$. This gives

$$G = \bigcup_{a \leq b} K \begin{pmatrix} \varpi_0^a & 0 \\ 0 & \varpi_b \end{pmatrix} K.$$

We have to prove this union is disjoint. That is, we have to recover the integers $a, b$ from the coset $KgK$, where

$$g = \begin{pmatrix} \varpi^a & 0 \\ 0 & \varpi^b \end{pmatrix}.$$

First, we have $a + b = \nu_F(\det h)$, for any $h \in KgK$. Next, the group index $(K : K \cap hKh^{-1})$ depends only on the coset $KhK$, and

$$(K : K \cap gKg^{-1}) = 1$$

if $b = a$, or

$$(K : K \cap gKg^{-1}) = (q + 1)q^{b-a-1}$$

if $b > a$, as we sought to show.

**Corollary 2.1.4: Countable Quotient**

If $K$ is a compact open subgroup of $G$, the set $G/K$ is countable.

***Proof*:**
As observed under box 1.2.28, it is enough to show that $G/K$ is countable for one choice of $K$: we take $K = GL_2(\mathfrak{o})$. The space $K\backslash G/K$ is certainly countable, and each double coset $KgK$ contains only finitely many cosets $g'K$.

We next define the Iwahori decomposition. First, the standard Iwahori subgroup of $\mathrm{GL}_2(F)$ is the compact open subgroup $I \leqslant G$:

$$I = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \ : \ a, d \in U_F,\ b \in \mathfrak{o},\ c \in \mathfrak{p} \right\}$$

Let $N' = N^w$ denote the subgroup of lower triangular unipotent matrices of $G$. Then:

**Proposition 2.1.5: Iwahori Decomposition**

If $I$ is the Iwahori subgroup, then:

$$I = (I \cap N')(I \cap T)(I \cap N)$$

More precisely, the product map

$$I \cap N' \times I \cap T \times I \cap N \longrightarrow I$$

is bijective, and a homeomorphism, for any ordering of the factors on the left hand side.

***Proof*:**
exercise.

This is useful for the following. Let $K = GL_2(\mathfrak{o})$; under the canonical surjection $K \to GL_2(k)$, the image of $I$ is the standard Borel subgroup of $GL_2(k)$. The Bruhat decomposition for $GL_2(k)$ implies

$$K = I \cup IwI.$$

Combining this with the Iwasawa decomposition for $G$, we obtain the more symmetric double-coset decomposition

$$G = BI \cup BwI = B(I \cap N') \cup Bw(I \cap N)$$

The cosets $BwI, BI$ are both open in $G$.

Furthermore, by letting $L = \mathfrak{o} \oplus \mathfrak{o}, L' = \mathfrak{o} \oplus \pi$, the Iwahori subgroup $I$ is then the common $G$-stabilizer of the two lattices $L, L'$.

## Haar Measures

A reminder that $F$ is a nonarchemedian field.

**Proposition 2.1.6: Smooth Functions Over F**

The vector-space $C_c^\infty(F)$ is spanned by indicator functions (characteristic functions) of sets $a + \mathfrak{p}^n$ for $a \in F$ and $n \in \mathbb{Z}$

***Proof*:**
Certainly the indicator function of $a + \mathfrak{p}^m$ lies in $C_c^\infty(F)$. Conversely, let $\varphi \in C_c^\infty(F)$. Since $\varphi$ has compact support, there exists $n \in \mathbb{Z}$ such that $\operatorname{supp} \varphi \subset \mathfrak{p}^n$. Also, $\varphi$ is fixed under translation by a compact open subgroup of $F$, hence by $\mathfrak{p}^m$, for some $m \in \mathbb{Z}$. Thus $\varphi$ is a linear combination of characteristic functions of sets $a + \mathfrak{p}^m$, $a \in \mathfrak{p}^n/\mathfrak{p}^m$.

Let us now define the Haar integral on our desired groups. Let $\mu$ be the Haar measure on $F$ (automatically left and right as $F$ is abelian), and let $\Phi_0$ denote the indicator function (or characteristic function) on $\mathfrak{o}$. Then as $\mathfrak{o}$ is compact:

$$\int_F \Phi_0(x) d\mu(x) = c_0 \qquad c \in \mathbb{R}_{>0}$$

If $\Phi_b$ is the indicator function on $a + \mathfrak{p}^b$ for $a \in F$ and $b \in \mathbb{Z}$, by translation invariance:

$$\int_F \Phi_b(x) d\mu(x) = c_0 q^{-b}$$

And as indicator functions span $C_c^\infty(F)$, we can use this to integrate any function.this integral has the following "homogeneity" property: if $\Phi \in C_c^\infty(F)$ and $y \in F^*$, then by the above equation we deduce:

$$\int_F \Phi(xy) d\mu(x) = \|y\|^{-1} \int_F \Phi(x) d\mu(x)$$

where $\|y\| = q^{\nu_F(y)}$. Thus, define the Haar measure $\mu^*$ on $F^*$ given by:

$$\mu^*(x) = \frac{\mu(x)}{\|x\|}$$

Then if $\Phi \in C_c^\infty(F^*)$, the function $x \mapsto \|x\|^{-1}\Phi(x)$ lies in $C_c^\infty(F)$ (and vanishes at 0), so:

$$\int_{F^\times} \Phi(x) d\mu^\times(x) = \int_F \Phi(x)\|x\|^{-1} d\mu(x), \qquad \text{for} \qquad \Phi \in C_c^\infty(F^\times).$$

and hence this certainly defines a Haar integral on $F^*$.

Now, viewing $M_2(F)$ under addition, we can define its Haar measure by interpreting it as the product of four copies of $F$, with the measure given by the tensor product of 4 copies of a Haar measure on $F$. Let $G = \mathrm{GL}_2(F)$. Then:

**Proposition 2.1.7: Haar measure on $\mathrm{GL}_2(F)$**

Let $\mu$ be a Haar measure on $M_2(F)$. For $\Phi \in C_c^\infty(\mathrm{GL}_2(F))$, the function $x \mapsto \Phi(x)\|\det x\|^{-2}$ (vanishing on $M_2(F) \setminus \mathrm{GL}_2(F)$) lies in $C_c^\infty(M_2(F))$. The functional

$$\Phi \mapsto \int_A \Phi(x)\|\det x\|^{-2}\, d\mu(x), \quad \Phi \in C_c^\infty(G),$$

is a left and right Haar integral on $\mathrm{GL}_2(F)$, i.e. $\mathrm{GL}_2(F)$ is unimodular.

***Proof*:**
Let $A = M_2(F)$ and $G = \mathrm{GL}_2(F)$. Take $g \in G$ and consider the functionals

$$\Phi \mapsto \begin{cases} \int_A \Phi(gx)\, d\mu(x), \\ \int_A \Phi(xg)\, d\mu(x), \end{cases} \qquad \Phi \in C_c^\infty(A).$$

Each is a Haar integral on $A$ and differs from the initial one by a positive constant (depending on $g$). To evaluate this constant, take $\Phi$ to be the indicator function of $\mathfrak{m} = M_2(\mathfrak{o})$. In the first instance, the function $x \mapsto \Phi(gx)$ is the characteristic function of the lattice $\mathfrak{m}' = g^{-1}\mathfrak{m}$. Thus

$$\int_A \Phi(gx)\, d\mu(x) = \mu(\mathfrak{m}') = \mu(\mathfrak{m}) \frac{[\mathfrak{m}' : \mathfrak{m} \cap \mathfrak{m}']}{[\mathfrak{m} : \mathfrak{m} \cap \mathfrak{m}']}$$

This quotient of indices depends only on the double coset $KgK$, $K = \mathrm{GL}_2(\mathfrak{o})$. Taking $g$ in diagonal form the Cartan decomposition (proposition 2.1.3), we get

$$\int_A \Phi(gx)\, d\mu(x) = \|\det g\|^{-2} \int_A \Phi(x)\, d\mu(x).$$

The second instance is treated in the same way to get

$$\int_A \Phi(xg)\, d\mu(x) = \|\det g\|^{-2} \int_A \Phi(x)\, d\mu(x).$$

The proposition then follows from simple manipulations. For example, if $\Phi \in C_c^\infty(G)$,

$$\begin{aligned} \int_A \Phi(xg)\|\det x\|^{-2}\, d\mu(x) &= \|\det g\|^{-2} \int_A \Phi(x)\|\det xg^{-1}\|^{-2}\, d\mu(x) \\ &= \int_A \Phi(x)\|\det x\|^{-2}\, d\mu(x), \end{aligned}$$

as we sought to show.

Having defined a Haar measure on $\mathrm{GL}_2(F)$, let us turn to the relevant subgroups. Let $N, T, B, Z$ be as in definition definition 1.5.1. The subgroup $N$, $Z$, and $T$, are taken care of as $N \cong F$, $Z \cong F^*$, and $T \cong F^* \times F^*$, leaving $B$. As $B \cong T \ltimes N$, define a linear functional on $C_c^\infty(B) := C_c^\infty(T) \otimes C_c^\infty(N)$ by:

$$\Phi \mapsto \int_T \int_N \Phi(tn) d\mu_T(t) d\mu_N(n)$$

where $\mu_T$, $\mu_N$ are the Haar measure of $T$ and $N$ respectively induced by the aforementioned isomorphisms. Certainly, this functional is left $B$-invariant, and hence a left-Haar integral on $B$. By the above functional, we may take $\mu_B := \mu_T \otimes \mu)N$. Importantly, this tensor product doesn't commute, taking into account the semidirect product defining $B$. Using $\mu_B$, we may write:

$$\Phi \mapsto \int_B \Phi(b) d\mu_B(b)$$

As the tensor product does not commute, $\mu_B$ is not unimodular, and hence there is a non-trivial module:

**Proposition 2.1.8: Module of Bruhat Group**

The module $\delta_B$ of the group $B$ is given by

$$\delta_B : tn \mapsto \|t_2/t_1\|, \quad n \in N, t = \begin{pmatrix} t_1 & 0 \\ 0 & t_2 \end{pmatrix} \in T. \tag{2.1}$$

***Proof*:**
Setting

$$c = sm, \quad m \in N, \quad s = \begin{pmatrix} s_1 & 0 \\ 0 & s_2 \end{pmatrix} \in T,$$

we get

$$\int_B \Phi(bc)\, d\mu_B(b) = \int_T \int_N \Phi(ts\, s^{-1} ns\, m)\, d\mu_T(t) d\mu_N(n).$$

We use the obvious isomorphism $N \to F$ to identify $\mu_N$ with a certain Haar measure $\mu_F$ on $F$. For $\varphi \in C_c^\infty(N)$, we then have

$$\begin{aligned} \int_N \varphi(s^{-1}ns)\, d\mu_N(n) &= \int_F \varphi\left(\begin{pmatrix} 1 & s_1^{-1} x s_2 \\ 0 & 1 \end{pmatrix}\right) d\mu_F(x) \\ &= \|s_1 s_2^{-1}\| \int_N \varphi(n)\, d\mu_N(n). \end{aligned}$$

By definition,

$$\int_B \Phi(bc)\, d\mu_B(b) = \delta_B(c)^{-1} \int_B \Phi(b)\, d\mu_B(b),$$

completing the proof.

This defines the desired semi-invariant measure:

**Corollary 2.1.9: semi-invariant measure on $\mathrm{GL}_2(\mathfrak{o})$**

The space $C_c^\infty(B\backslash G, \delta_B^{-1})$ admits a positive semi-invariant measure $\dot{\mu}$, where $\delta_B$ is given by proposition 2.1.8. If $K = \mathrm{GL}_2(\mathfrak{o})$, there is a Haar measure $\mu_K$ on $K$ such that

$$\int_{B\backslash G} f(g)\, d\dot{\mu}(g) = \int_K f(k)\, d\mu_K(k),$$

for $f \in C_c^\infty(B\backslash G, \delta_B^{-1})$.

***Proof*:**
The character $\delta_B$ is trivial on the compact group $K \cap B$. Restriction of functions is an isomorphism $C_c^\infty(B\backslash G, \delta_B^{-1}) \to C_c^\infty(K \cap B\backslash K, 1)$, where 1 denotes the trivial character of $K \cap B$. The semi-invariant measure $\dot{\mu}$ thus restricts to a semi-invariant measure on $C_c^\infty(K \cap B\backslash K, 1)$, but so does any Haar measure on $K$.

Note that $\mu_K$ is effectively just the restriction of a Haar measure $\mu_G$ on $G$. Using this, we see there is a left Haar measure $\mu_B$ on $B$ such that

$$\int_G \varphi(g)\, d\mu_G(g) = \int_K \int_B \varphi(bk)\, d\mu_B(b) d\mu_G(k), \quad \varphi \in C_c^\infty(G). \tag{2.2}$$

showing that the measure and integral "commute" with the decomposition $G = BK$.

## 2.2 Representation of the Mirabolic Group

A lot of the representation of $\mathrm{GL}_2(F)$ can be simplified by first finding the representation of the following subgroup:

**Definition 2.2.1: Mirabolic Subgroup**

The group $M \subseteq \mathrm{GL}_2(F) = G$ given by:

$$M = \left\{ \begin{pmatrix} a & x \\ 0 & 1 \end{pmatrix} \ : \ a \in F^*,\ x \in F \right\}$$

is called the *mirabolic subgroup* of $G$. Letting $S = T \cap M \cong F^*$, then $M \cong S \ltimes N$.

Let us start by analyzing the smooth representations of $N$ from which we shall do induction. Let $(\pi, V)$ be a smooth representation and $\vartheta$ a character of $N$. Let $V(\vartheta)$ be the subspace generated by all $\pi(n)v - \vartheta(n)v$. This is an $N$ subspace, and $V_\vartheta = V/V(\vartheta)$ is well-defined and, by construction, as the maximal $N$-quotient where the action of $N$ acts via the character $\vartheta$. If $\vartheta$ is trivial, then $V(\vartheta) = V(N)$ and so $V_\vartheta = V_N$.

From this, we can define the following lemma that can be thought of as the generalization of the test in Fourier analysis where $\int f e^{2i\pi k} dx = 0$:

**Lemma 2.2.2: Generalized Harmonic Decomposition**

Let $\mu_N$ be a Haar measure on $N$ and $\vartheta$ a character of $N$.

1. Let $(\pi, V)$ be a smooth representation of $N$ and $v \in V$. The vector $v$ lies in $V(\vartheta)$ if and only if there is a compact open subgroup $N_0$ of $N$ such that

$$\int_{N_0} \vartheta(n)^{-1} \pi(n) v \, d\mu_N(n) = 0 \tag{2.3}$$

2. The process $(\pi, V) \mapsto V_\vartheta$ is an exact functor from $\mathrm{Rep}(N)$ to the category of complex vector spaces.

Think of the integral as the generalized projection operator, $\int_{N_0} \vartheta(n)^{-1}\pi(n)v\, d\mu_N(n) = 0$. Multiplying by $\vartheta(n)^{-1}$ is replaces the $e^{-2\pi i k}$.

***Proof*:**

1. First assume that $\vartheta$ is trivial on the trivial character. The group $N \cong F$ is the union of an ascending sequence of compact open subgroups. So, if

$$v = \sum_{i=1}^{r} v_i - \pi(n_i)v_i \in V(N),$$

there is a compact open subgroup $N_0$ of $N$ containing all $n_i$. Then the relation given by equation 2.3 holds for this choice of $N_0$.

Conversely, let $v \in V$ and suppose the integral relation holds. Then there is an open normal subgroup $N_1$ of $N_0$ such that $v \in V^{N_1}$. The space $V^{N_1}$ carries a representation of the finite group $N_0/N_1$. Therefore, in the obvious notation, $V^{N_1} = V^{N_1}(N_0/N_1) \oplus V^{N_0}$ and the map

$$w \mapsto \mu_N(N_0)^{-1} \int_{N_0} \pi(n)w \, d\mu_N(n), \quad w \in V^{N_1},$$

is the $N_0$-projection $V^{N_1} \to V^{N_0}$. This has kernel $V^{N_1}(N_0/N_1) \subset V(N)$. But then this proves (1) for the trivial character of $N$.

Now let $\vartheta$ be an arbitrary character of $N$, and consider the representation $(\pi', V')$ of $N$, where $V' = V$ and $\pi'(n) = \vartheta(n)^{-1}\pi(n)$. Then $V(\vartheta) = V'(N)$ and so (1) follows in general.

2. right-exactness is clear as we are taking the quotient. For the other way, if $f : U \to V$ is injective we want that the induced $\overline{f}$ is injective. If $\overline{f}(v) = 0$, then for injectivity it must be in $V(\vartheta)$. As it is equal to $\overline{0}$ we get:

$$\int_{N_0} \vartheta(n)^{-1}\pi(n) \cdot (f(v)) \, d\mu_N(n) = 0$$

As $f$ is an $N$-homomorphism, it commutes with the action given by $\pi$

$$\int_{N_0} \vartheta(n)^{-1} f(\pi(n) \cdot v) \, d\mu_N(n) = 0$$

and by linearity $f$ comes out, and by injectivity of $f$ we can drop $f$, and finally by part 1 we get that $v \in V(\vartheta)$ giving injectivity of $\overline{f}$, completing the proof.

Some important remarks. If $(\pi, V)$ is a smooth representation of $N$ (or of $M$), then $V(N)$ is an $N$-(or $M$-) subspace of $V$. The exact sequence

$$0 \to V(N) \to V \to V_N \to 0$$

gives an exact sequence

$$0 \to V(N)_N \to V_N \to V_N \to 0$$

in which the map $V_N \to V_N$ is the identity. Therefore

$$V(N)_N = 0 \quad \text{and} \quad V(N)(N) = V(N)$$

If $\vartheta \neq 1$, as $N$ acts trivially on $V/V(N)$, we have $(V/V(N))_\vartheta = 0$ and so the inclusion $V(N) \to V$ induces an isomorphism

$$V(N)_\vartheta \cong V_\vartheta.$$

**Proposition 2.2.3: Character Invariant Subspace Never Whole Space**

Let $(\pi, V)$ be a smooth representation of $N$ and let $0 \neq v \in V$. Then there exists a character $\vartheta$ such that $v \notin V(\vartheta)$.

***Proof*:**
We write
$$N_j = \begin{pmatrix} 1 & \mathfrak{p}^j \\ 0 & 1 \end{pmatrix}, \quad j \in \mathbb{Z}.$$
Take $v \in V$, $v \neq 0$. We choose $j_0 \in \mathbb{Z}$ such that $N_{j_0}$ fixes $v$. For $j \leq j_0$, let $V_j$ denote the $N_j$-space generated by $v$. This is the direct sum of isotypic components $V_j^{\eta_j}$, as $\eta$ ranges over the characters of $N_j$ trivial on $N_{j_0}$. For each $j \leq j_0$, there exists $\eta_j$ such that $V_j^{\eta_j} \neq 0$; by definition, we have
$$\int_{N_j} \eta_j(n)^{-1}\pi(n)v\, dn \neq 0.$$
The $N_{j-1}$-space generated by $V_j^{\eta_j}$ is contained in $V_{j-1}$, so we may choose $\eta_{j-1}$ such that $\eta_{j-1}|_{N_j} = \eta_j$. This satisfies the necessary properties to take the limit of these characters, and as $N$ is a limit of the $N_j$ we get that there exists a character $\vartheta$ of $N$ such that, for all $j \leq j_0$, we have
$$\int_{N_j} \vartheta(n)^{-1}\pi(n)v\, dn \neq 0.$$
Therefore $v \notin V(\vartheta)$, as required.

**Corollary 2.2.4: Trivial If Trivial On Characters**

Let $(\pi, V)$ be a smooth representation of $N$. If $V_\vartheta =$ for all characters $\vartheta$ of $N$, then $V = 0$

***Proof*:**
immediate

Now, let $(\pi, V)$ be a smooth representation of $M$. Then we can deduce:

**Corollary 2.2.5: Trivial Representation of Mirabolic Group**

Let $(\pi, V)$ be a smooth representation of $M$. Suppose $V_N = 0$ and for some non-trivial character $\vartheta$ of $N$, $V_\vartheta = 0$. Then $V = 0$.

***Proof*:**
Now let $(\pi, V)$ be a smooth representation of $M$. The space $V(N)$ is then an $M$-subspace of $V$ and $V_N$ carries a natural representation of $M/N = S$: On the other hand, $\pi(S)$ permutes transitively the subspaces $V(\vartheta)$, $\vartheta \neq 1$: for $s \in S$, $\pi(s) \cdot V(\vartheta) = V(\vartheta')$, where $\vartheta'(n) = \vartheta(s^{-1}ns)$. The corollary follows immediately.

Now, fix a non-trivial character $\vartheta$ of $N$. We start by looking at $\mathrm{Ind}_N^M \vartheta$ and c-$\mathrm{Ind}_N^M \vartheta$. As $N \trianglelefteq M$

and all non-trivial characters $\vartheta, \vartheta'$ of $N$ are conjugate, the inducted and compacted inducted representations are $M$-isomorphic.

**Theorem 2.2.6: c-$\mathrm{Ind}_N^M \vartheta$ is Irreducible**

Let $\vartheta$ be a non-trivial character of $N$ and set $\mathcal{W} = \mathrm{Ind}_N^M \vartheta$, $\mathcal{W}^c = \text{c-}\mathrm{Ind}_N^M \vartheta$. Let $\alpha : \mathcal{W} \to \mathbb{C}$ denote the canonical map $f \mapsto f(1)$.

1. We have $\mathcal{W}(N) = \mathcal{W}^c(N) = \mathcal{W}^c$ and $(\mathcal{W}/\mathcal{W}^c)(N) = 0$.
2. The map $\alpha$ induces isomorphisms $\mathcal{W}_\vartheta \cong \mathbb{C}$ and $\mathcal{W}^c_\vartheta \cong \mathbb{C}$.

As a consequence, c-$\mathrm{Ind}_N^M \vartheta$ is irreducible over $M$.

***Proof*:**

1. Let $f \in \mathcal{W}$ and $n \in N$. For $a \in S$, we have $f(an) = \vartheta(ana^{-1})f(a)$. As there is an integer $j$ such that $N_j$ fixes $f$ by proposition 2.2.3, $f(a) = 0$ if $\|\det a\|$ is sufficiently large. On the other hand, $f(an) = f(a)$ if $\|\det a\|$ is sufficiently small, so $nf - f$ vanishes at $a$ if $\|\det a\|$ is sufficiently small. This implies that $nf - f \in \mathcal{W}^c$. Thus $\mathcal{W}(N) \subset \mathcal{W}^c$ and $N$ acts trivially on $\mathcal{W}/\mathcal{W}^c$.

   Next, to prove that $\mathcal{W}^c(N) = \mathcal{W}^c$, let $\psi$ be the character of $F$ defined by
   $$\psi(x) = \vartheta\left(\begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}\right).$$
   Take $a \in F^\times$, $j \in \mathbb{Z}_{\geq 1}$. Let $f_{a,j} \in \mathcal{W}^c$ be the function such that
   $$f_{a,j} : \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}\begin{pmatrix} au & 0 \\ 0 & 1 \end{pmatrix} \mapsto \psi(x), \quad u \in U_F^j,$$
   and which vanishes elsewhere. By similar augments as done before, the functions $f_{a,j}$ span $\mathcal{W}^c$ over $\mathbb{C}$. We have
   $$\begin{pmatrix} 1 & z \\ 0 & 1 \end{pmatrix} f_{a,j}\left(\begin{pmatrix} b & 0 \\ 0 & 1 \end{pmatrix}\right) = \psi(bz) f_{a,j}\left(\begin{pmatrix} b & 0 \\ 0 & 1 \end{pmatrix}\right).$$
   We deduce that $nf_{a,j}$ has the same support as $f_{a,j}$, $n \in N$. We can certainly find $x \in F$ such that the function $u \mapsto \psi(aux)$, $u \in U_F^j$, is constant, equal to $c$ say, with $c \neq 1$. If $n = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}$, then $nf_{a,j} - f_{a,j} = (c-1)f_{a,j}$, whence $f_{a,j} \in \mathcal{W}^c(N)$ and so $\mathcal{W}^c(N) = \mathcal{W}^c$. This implies $\mathcal{W}(N) = \mathcal{W}^c$.
2. The map $\alpha$ induces a surjection $\mathcal{W}_\vartheta \to \mathbb{C}$. On the other hand, since $N$ acts trivially on $\mathcal{W}/\mathcal{W}^c$, the inclusion $\mathcal{W}^c \to \mathcal{W}$ induces an isomorphism $\mathcal{W}^c_\vartheta \cong \mathcal{W}_\vartheta$. To prove (2), therefore, it is enough to show that any $f \in \mathcal{W}^c$ with $f(1) = 0$ belongs to $\mathcal{W}^c(\vartheta)$. A function $f$, vanishing at 1, is a finite linear combination of functions $f_{a,j}$ with $a \notin U_F^j$, so it is enough to treat such functions. However, as $a \notin U_F^j$, there exists $x \in F$ such that the function $u \mapsto \psi(aux) - \psi(x)$, $u \in U_F^j$, is a non-zero constant. Taking $n = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}$, the same

calculation as before shows that $nf_{a,j} - \vartheta(n)f_{a,j}$ is a non-zero constant multiple of $f_{a,j}$, whence $f_{a,j} \in \mathcal{W}^c(\vartheta)$ and $\mathcal{W}^c(\vartheta) = \mathcal{W}^c$

Finally, for the irreducibility, let $V$ be an $M$-subspace of $\mathcal{W}^c$. As $\mathcal{W}^c_N = 0$, the spaces $V_N, (\mathcal{W}^c/V)_N$ are both zero. The sequence

$$0 \to V_\vartheta \to \mathcal{W}^c_\vartheta \to (\mathcal{W}^c/V)_\vartheta \to 0$$

is exact. As $\dim \mathcal{W}^c_\vartheta = 1$, we conclude that $\dim V_\vartheta$ is 0 or 1. In the first case, $V = 0$ by corollary 2.2.4. In the second, $(\mathcal{W}^c/V)_\vartheta = 0$ and so $\mathcal{W}^c = V$, showing it is irreducible and completing the proof.

The following from the textbook [2] summarizes the key points:

1. A function $f \in W$ is determined by its restriction to $S \cong F^\times$. The restriction $f|_{F^\times}$ is a smooth function on $F^\times$.

2. A smooth function $\varphi$ on $F^\times$ is of the form $\varphi = f|_{F^\times}$, for some $f \in W$, if and only if there exists $c > 0$ such that $\varphi(x) = 0$ for all $x$ satisfying $\|x\| > c$.

3. A function $f \in W$ lies in $W^c$ if and only if $f|_{F^\times} \in C^\infty_c(F^\times)$.

**2.2.7 Lack of Irreducibility of Dual** The above shows the representation $\operatorname{Ind}_N^M \vartheta$ is never irreducible, for any non-trivial character $\vartheta$ of $N$. The Duality Theorem (theorem 1.3.12) implies

$$(\text{c-Ind}_N^M \vartheta)^\vee \cong \operatorname{Ind}_N^M \check{\vartheta}.$$

Thus $\text{c-Ind}_N^M \vartheta$ provides an example of an irreducible smooth representation with reducible contragredient.

We can now finish classifying *all* irreducible representations of the mirabolic group $M$. The following theorem is the key technical tool to show determine we found them all. As before let $\vartheta$ be a non-trivial character of $N$. Let $(\pi, V)$ be a smooth representation of $M$. Frobenius Reciprocity gives a canonical isomorphism

$$\operatorname{Hom}_N(V, V_\vartheta) \cong \operatorname{Hom}_M(V, \operatorname{Ind}_N^M V_\vartheta).$$

Let $q : V \to V_\vartheta$ denote the quotient map and let $q_\star$ be the map $V \to \operatorname{Ind}_N^M V_\vartheta$ corresponding to $q$ under this isomorphism: for $v \in V$,

$$q_\star(v) = (m \mapsto q\,(\pi(m)v))$$

**Theorem 2.2.8: Invariant Space equivalent to $\text{c-Ind}_N^M V_\vartheta$**

Let $(\pi, V)$ be a smooth representation of $M$. The $M$-homomorphism $q_\star : V \to \operatorname{Ind}_N^M V_\vartheta$ induces an isomorphism $V(N) \cong \text{c-Ind}_N^M V_\vartheta$.

***Proof*:**
The $N$-space $V_\vartheta$ is a direct sum of copies of $\vartheta$. Therefore $\mathrm{Ind}_N^M V_\vartheta$ is a direct sum of copies of $\mathrm{Ind}_N^M \vartheta$. By theorem 2.2.6

$$(\mathrm{Ind}_N^M V_\vartheta)(N) = \text{c-Ind}_N^M V_\vartheta = (\text{c-Ind}_N^M V_\vartheta)(N).$$

The $M$-homomorphism $q_\star : V \to \mathrm{Ind}_N^M V_\vartheta$ surely maps $V(N)$ to $(\mathrm{Ind}_N^M V_\vartheta)(N) = \text{c-Ind}_N^M V_\vartheta$.

Let $W = \operatorname{Ker} q_\star \cap V(N)$ and $C = \text{c-Ind}_N^M V_\vartheta / q_\star(V(N))$. By lemma 2.2.2 the natural map $W_N \to V(N)_N$ is injective, so $W_N = 0$. Likewise, $(\text{c-Ind}\, V_\vartheta)_N$ is zero, so $C_N = 0$.

Now, the map $q_\star : V \to \text{c-Ind}\, V_\vartheta$ induces a map

$$q_{\star,\vartheta} : V_\vartheta = V(N)_\vartheta \to (\text{c-Ind}\, V_\vartheta)_\vartheta.$$

By theorem 2.2.6(2), the canonical $N$-map $\mathrm{Ind}\, V_\vartheta \to V_\vartheta$ induces an isomorphism $\alpha_\vartheta : (\text{c-Ind}\, V_\vartheta)_\vartheta \to V_\vartheta$. The composite map $\alpha_\vartheta \circ q_{\star,\vartheta} : V_\vartheta \to V_\vartheta$ is the identity. However, the kernel of this map is $W_\vartheta$ and its cokernel is isomorphic to $C_\vartheta$. As the spaces $W_N, W_\vartheta, C_N, C_\vartheta$ are all zero, by corollary 2.2.5 both $W$ and $C$ are trivial and $q_\star : V(N) \to \text{c-Ind}\, V_\vartheta$ is an isomorphism, completing the proof.

**Theorem 2.2.9: Classification of Irreducible Representation of Mirabolic Group**

Let $(\pi, V)$ be an irreducible smooth representation of $M$. Either:

1. $\dim V = 1$ and $\pi$ is the inflation of a character of $M/N \cong F^\times$
2. $\dim V$ is infinite and $\pi \cong \text{c-Ind}_N^M \vartheta$, for any character $\vartheta \neq 1$ of $N$.

In case (1), $\dim V_N = 1$ and $V_\vartheta = 0$ for $\vartheta \neq 1$. In case (2), $V_N = 0$ and $\dim V_\vartheta = 1$ for all $\vartheta \neq 1$.

***Proof*:**
If $V(N) = 0$, then $N$ acts trivially on $V$. The group $M/N$ is abelian, so Schur's Lemma 2.6 implies $\dim V = 1$ and we are in case (1).

If $V(N) \neq 0$, then $V(N) = V$ and $V_N = 0$. Therefore $V_\vartheta \neq 0$ for all characters $\vartheta \neq 1$ of $N$, and so $\dim V$ is infinite. Theorem 2.2.8 implies that $V = V(N)$ is $M$-isomorphic to $\text{c-Ind}_N^M V_\vartheta$. The $N$-space $V_\vartheta$ is a direct sum of copies of $\vartheta$, so $V$ is a direct sum of copies of $\text{c-Ind}_N^M \vartheta$ and, since it is irreducible, $V \cong \text{c-Ind}_N^M \vartheta$.

## 2.3 Representation of $\mathrm{GL}_2(F)$: Jacquet Modules and Induced Representations

We now classify the irreducible smooth representations of $\mathrm{GL}_2(F) = G$. In this section, we deal completely with *principal series representations*, that is irreducible representations where $V_N \neq 0$: This is the equivalent of looking to see if $\pi$ contains the trivial character of $N$. The terminology comes from functional analysis when taking the spectrum of a topological group with the continuous spectrum associated to the principal (as in main part) series of characters and discrete spectrum's will be later associated to the cuspidal representations. It shall be the case that much of the important representation theory on the Galois side will be in relation to the cuspidal, but that cannot be properly studied without first grasping the whole picture of the representation of $\mathrm{GL}_2(F)$.

**Definition 2.3.1: Principal Series Representation**

Let $(\pi, V)$ be an irreducible representations of $\mathrm{GL}_n(F)$ (notice $n \in \mathbb{N}_{>0}$). Then it is called a *principal series* if $V_N \neq 0$.

When $V_N = 0$ these shall be called *cuspidal representations* or *supercuspidal representations*, which could be thought of as the "singular" case (and a singularity in the form of a cusp does appear when looking at the geometric picture modular forms; see ref:HERE), and are dealt with in ref:HERE.

Let $(\pi, V)$ be a smooth representation of $\mathrm{GL}_2(F)$ and $V(N)$ denote the subspace of $V$ spanned by the vectors $v - \pi(x)v$, for $v \in V$ and $x \in N$. The space $V_N = V/V(N)$ inherits a representation $\pi_N$ of $B/N = T$, which is smooth. This representation is given a name:

**Definition 2.3.2: Jacquet Module**

The representation $(\pi_N, V_N)$ is called the *Jacquet module* of $(\pi, V)$ at $N$.
This gives the *Jacquet functor*

$$\begin{aligned} \mathrm{Rep}(G) &\to \mathrm{Rep}(T), \\ (\pi, V) &\mapsto (\pi_N, V_N), \end{aligned}$$

which is exact and additive.

Let $(\sigma, W)$ be a smooth representation of $T$. View $\sigma$ as a smooth representation of $B$ which is trivial on $N$, and form the smooth induced representation $\mathrm{Ind}_B^G \sigma$. We shall often abbreviate $\mathrm{Ind}_B^G \sigma = \mathrm{Ind}\, \sigma$, since $B$ and $G$ are the only groups involved for most of this section. This is often called *parabolic induction*.

If $(\pi, V)$ is a smooth representation of $G$, Frobenius Reciprocity gives an isomorphism

$$\mathrm{Hom}_G(\pi, \mathrm{Ind}\, \sigma) \cong \mathrm{Hom}_B(\pi, \sigma).$$

However, $\sigma$ is trivial on $N$ so any $B$-homomorphism $\pi \to \sigma$ factors through the quotient map $\pi \to \pi_N$. Thus:

$$\mathrm{Hom}_G(\pi, \mathrm{Ind}\, \sigma) \cong \mathrm{Hom}_T(\pi_N, \sigma).$$

**Proposition 2.3.3: Jaquet Module and Principal Series Representation**

Let $(\pi, V)$ be an irreducible smooth representation of $G$. The following are equivalent:

1. The Jacquet module $V_N$ is non-zero.
2. The representation $\pi$ is isomorphic to a $G$-subspace of a representation $\mathrm{Ind}_B^G \chi$, for some character $\chi$ of $T$.

***Proof*:**

(1)⇒(2) We have to show that the Jacquet module $V_N$ admits an irreducible $T$- (or $B$-) quotient.

Choose $v \in V, v \neq 0$. Since $V$ is irreducible over $G$, any element of $V$ is a finite linear combination of translates $\pi(g)v$ of $v$, for various $g \in G$. Write $K = \mathrm{GL}_2(\mathfrak{o})$. The vector $v$ is fixed by a subgroup $K'$ of $K$ of finite index; let $\{v_1, v_2, \ldots, v_r\}$ be the distinct elements of the form $\pi(k)v, k \in K$. In particular, $r \leq (K : K')$. Since $G = BK$, the elements $v_1, \ldots, v_r$ generate $V$ over $B$, and their images generate $V_N$ over $T$.

Thus $V_N$ is finitely generated as a representation of $T$. We choose a minimal generating set $\{u_1, \ldots, u_t\}, t \geq 1$, say. A standard Zorn's Lemma argument shows that $V_N$ has a $T$-subspace $U$, containing $u_1, \ldots, u_{t-1}$, and maximal for the property $u_t \notin U$. Then $U$ is a maximal $T$-subspace of $V_N$ and $V_N/U$ is an irreducible representation of $T$, hence a character .

(2)⇒(1) As $\mathrm{Hom}_G(\pi, \mathrm{Ind}\,\sigma) \cong \mathrm{Hom}_T(\pi_N, \sigma)$. we get

$$\mathrm{Hom}_T(\pi_N, \chi) \cong \mathrm{Hom}_G(\pi, \mathrm{Ind}\,\chi) \neq 0,$$

so $\pi_N \neq 0$.

Next, let $\mu_N$ be a Haar measure on $N$ and let $t \in T$. The measure $S \mapsto \mu_N(t^{-1}St)$ is the Haar measure $\delta_B(t)\mu_N$:

$$\int_N f(txt^{-1})\,d\mu_N(x) = \delta_B(t) \int_N f(x)\,d\mu_N(x), \quad f \in C_c^\infty(N).$$

As before, let $w$ denote the permutation matrix

$$w = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

If $\sigma$ is a smooth representation of $T$, we can form the representation $\sigma^w : t \mapsto \sigma(wtw^{-1})$, and view it as a representation of $B$ which is trivial on $N$.

Let $\alpha_\sigma$ denote the canonical $B$-map $\mathrm{Ind}\,\sigma \to \sigma$ given by $f \mapsto f(1)$. It induces a canonical $T$-map $(\mathrm{Ind}\,\sigma)_N \to \sigma$, which we continue to denote $\alpha_\sigma$.

**Theorem 2.3.4: Restriction-induction Lemma**

Let $(\sigma, U)$ be a smooth representation of $T$ and set $(\Sigma, X) = \mathrm{Ind}_B^G \sigma$. There is an exact sequence of representations of $T$:

$$0 \to \sigma^w \otimes \delta_B^{-1} \to \Sigma_N \xrightarrow{\alpha_\sigma} \sigma \to 0.$$

First, a technical lemma:

**Lemma 2.3.5: Technical Lemma**

Recall that $G = B \cup BwN$. A function $f \in X$ lies in $V$ if and only if $\operatorname{supp} f \subset BwN$, that is if $f \in X$ then $f \in V$ if and only if there is a compact open subgroup $N_0$ of $N$ (depending on $f$) such that $\operatorname{supp} f \subset BwN_0$.

***Proof*:**
A function $f \in X$ lies in $V$ if and only if $f(1) = 0$. Since $f$ is $G$-smooth, such a function $f$ vanishes on a set $BN_0'$, where $N_0'$ is some compact open subgroup of $N'$. The identity (for $x \neq 0$)

$$\begin{pmatrix} 1 & 0 \\ x & 1 \end{pmatrix} = \begin{pmatrix} 1 & x^{-1} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} -x^{-1} & 0 \\ 0 & x \end{pmatrix} w \begin{pmatrix} 1 & x^{-1} \\ 0 & 1 \end{pmatrix} \in Bw \begin{pmatrix} 1 & x^{-1} \\ 0 & 1 \end{pmatrix}$$

implies that $\operatorname{supp} f \subset BwN_0$, for some compact open subgroup $N_0$ of $N$, as required.

***Proof*:**
(of theorem 2.3.4)

By definition, $X$ is the space of $G$-smooth functions $f : G \to U$ such that $f(bg) = \sigma(b)f(g)$, $b \in B, g \in G$. The canonical map $\alpha_\sigma : X \to U$ amounts to restriction of functions to $B$. Set $V = \operatorname{Ker} \alpha_\sigma$. Thus $V$ provides a smooth representation of $B$ and there is an exact sequence

$$0 \to V_N \to X_N \to U \to 0.$$

We have to identify the $T$-representation $V_N$.

Let $f \in V$; in view of the preceding lemma, we can define a function $f_N : T \to U$ by

$$f_N(x) = \int_N f(xwn)\, dn = \sigma(x) f_N(1), \quad x \in T.$$

By 8.1 Lemma, the kernel of the map $f \mapsto f_N$ is $V(N)$, and so $f \mapsto f_N(1)$ gives a bijective map $V_N \to U$. Taking $t \in T$ and $f \in V$, we have

$$\begin{aligned} (tf)_N(x) &= \int_N f(xwnt)\, dn \\ &= \delta_B(t^{-1}) \int_N f(xt^w wn)\, dn = \delta_B^{-1}(t)(t^w f_N)(x). \end{aligned}$$

Thus $f \mapsto f_N(1)$ is a $B$-homomorphism $V \to \sigma^w \otimes \delta_B^{-1}$ inducing a $T$-isomorphism $V_N \cong \sigma^w \otimes \delta_B^{-1}$.

**Proposition 2.3.6: Non-cuspidal Irreducible Representation Are Admissible**

Let $(\pi, V)$ be an irreducible smooth representation of $\mathrm{GL}_2(F)$ which is not cuspidal. The representation $\pi$ is admissible.

***Proof*:**

By definition, $V_N \neq 0$. By proposition 2.3.3, $\pi$ is equivalent to a sub-representation of $\mathrm{Ind}_B^G\chi$, for some character $\chi$ of $T$. It is enough, therefore, to prove that $\mathrm{Ind}\chi$ is admissible.

We fix a compact open subgroup $K$ of $G$; shrinking it if necessary, we may assume that $K \subset K_0 = \mathrm{GL}_2(\mathfrak{o})$. The space $X^K$ of $K$-fixed points in $\mathrm{Ind}\chi$ consists of the functions $f : G \to \mathbb{C}$ satisfying

$$f(bgk) = \chi(b)f(g), \quad b \in B, g \in G, k \in K. \tag{2.4}$$

We have $G = BK_0$, so the set $B\backslash G/K$ is finite, and each double coset $BgK$ supports, at most, a one-dimensional space of functions satisfying equation 2.4. Thus $X^K$ is finite-dimensional, as we sought to show.

Note that irreducible *cuspidal* representations of $\mathrm{GL}_2(F)$ are likewise admissible, and hence all irreducible representations of $\mathrm{GL}_2(F)$ are admissible (see ref:HERE for the cuspidal equivalent, corollary 10.2 in [2]). We shall now find a more granular classification of principal series. First recall from example 1.2.8 that smooth characters on $\mathrm{GL}_n(F)$ factored via the determinant and a character of $F^*$. Define the following:

**Definition 2.3.7: Twist On Representation**

Let $(\pi, V)$ is a smooth representation of $G$ and $\varphi$ a character of $F^*$. A *twist* $\varphi\pi$ is a smooth representation $(\varphi\pi, V)$ of $G$ given by:

$$\varphi\pi(g) = \varphi(\det g)\pi(g), \quad g \in G$$

Similarly for characters of $T$: if $\chi = \chi_1 \otimes \chi_2$ is a character of $T$ and $\varphi$ is a character of $F^\times$, put $\varphi\cdot\chi = \varphi\chi_1 \otimes \varphi\chi_2$. If we inflate $\varphi\cdot\chi$ to a representation of $B$ trivial on $N$, we get $\varphi\cdot\chi = (\varphi \circ \det|_B) \otimes \chi$. It follows immediately that

$$\mathrm{Ind}_B^G(\varphi\cdot\chi) \cong \varphi\mathrm{Ind}_B^G\chi$$

This allows us to make convenient adjustments to the character $\chi$ without changing the essential structure of the induced representation. With this definition, we make precise the structure of representations of the form $\mathrm{Ind}_B^G\chi$. The main step is:

**Theorem 2.3.8: Irreducibility Criterion**

Let $\chi = \chi_1 \otimes \chi_2$ be a character of $T$, and set $(\Sigma, X) = \mathrm{Ind}_B^G\chi$.

1. The representation $(\Sigma, X)$ is reducible if and only if $\chi_1\chi_2^{-1}$ is either the trivial character or the character $x \mapsto \|x\|^2$ of $F^\times$.
2. Suppose that $(\Sigma, X)$ is reducible. Then:
   (a) the $G$-composition length of $X$ is 2;
   (b) one composition factor of $X$ has dimension 1, the other is of infinite dimension;
   (c) $X$ has a 1-dimensional $G$-subspace if and only if $\chi_1\chi_2^{-1} = 1$;
   (d) $X$ has a 1-dimensional $G$-quotient if and only if $\chi_1\chi_2^{-1}(x) = \|x\|^2$, $x \in F^\times$.

We will refine this to a classification of the irreducible principal series representations

Let:

$$V = \{f \in X : f(1) = 0\}.$$

This is a $B$-subspace of $X$ and we have an exact sequence

$$0 \to V \to X \to C \to 0,$$

where the one-dimensional space $C = X/V$ carries the character $\chi$ of $T$. By the Restriction-Induction Lemma theorem 2.3.4, so $V_N \cong \delta_B^{-1}\chi^w$.

**Proposition 2.3.9: Irreducible Kernel**

Let $W$ be the kernel $V(N)$ of the canonical map $V \to V_N$. The space $W$ is irreducible over $B$.

First an important technical lemma:

**Lemma 2.3.10: Representation of V**

For $f \in V$, define a function $f_N \in C_c^\infty(N)$ by $f_N(n) = f(wn)$, $n \in N$. The map

$$\begin{aligned} V &\longrightarrow C_c^\infty(N), \\ f &\longmapsto f_N \end{aligned}$$

is an $N$-isomorphism.

***Proof*:**
As in theorem 2.3.4, the support of $f \in V$ is contained in $Bw\mathcal{N}$, for some compact open subgroup $\mathcal{N}$ of $N$. The assertions follow immediately (observing that the notation $f_N$ here is not the same as that in theorem 2.3.4).

***Proof*:**
(of proposition 2.3.9)

We shall prove that $W$ is irreducible as a representation of the mirabolic group $M$: recall under lemma 2.2.2 it was shown that $W_N = 0$ and $W = W(N)$. For $\varphi \in C_c^\infty(N)$ and $a \in F^\times$, we define $a\varphi \in C_c^\infty(N)$ by

$$a\varphi \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} = \chi_2(a)\varphi \begin{pmatrix} 1 & a^{-1}x \\ 0 & 1 \end{pmatrix}.$$

This gives an action of $F^\times$ on $C_c^\infty(N)$ which we regard as an action of the group $S$ of matrices $\begin{pmatrix} a & 0 \\ 0 & 1 \end{pmatrix}$. We combine this with the natural action of $N$ to give $C_c^\infty(N)$ the structure of a smooth representation of $M$. With this structure, the map in lemma 2.3.10 is an $M$-isomorphism is an $M$-isomorphism.

Let $\vartheta$ be a non-trivial character of $N$. The map $f \mapsto \vartheta f$ is a linear automorphism of $V = C_c^\infty(N)$ carrying $V(N)$ to $V(\vartheta)$. Since $V/V(N)$ is one dimensional, so is $\dim V_\vartheta = 1$. However, since $N$ acts trivially on $V_N$, the inclusion $W \to V$ induces an isomorphism $W_\vartheta \cong V_\vartheta$, hence $W_\vartheta \cong \vartheta$. Then apply theorem 2.2.9 to get $W = W(N) \cong \text{c-Ind}_N^M \vartheta$ which, by corollary 2.2.5 is irreducible.

As a direct consequence of the Proposition, we have:

**Corollary 2.3.11: Length of Induced Representation by Character**

As a representation of $B$ or of $M$, $\mathrm{Ind}_B^G\chi$ has composition length 3. Two of the composition factors have dimension one, and the third is of infinite dimension. In particular, the $G$-composition length of the representation $\mathrm{Ind}_B^G\chi$ is at most 3.

**Proposition 2.3.12: Detecting one-dimensional Subrepresentations**

The following are equivalent:

($\Rightarrow$) (1) $\chi_1 = \chi_2$

($\Rightarrow$) (2) $X$ has a one-dimensional $N$-subspace

When these conditions hold,

1. $X$ has a unique one-dimensional $N$-subspace $X_0$;
2. $X_0$ is a $G$-subspace of $X$, and it is not contained in $V$.

***Proof*:**
If (1) holds, by twisting we may as well take $\chi_1 = \chi_2 = 1$. The (non-zero) constant function then spans a one-dimensional $G$-subspace of $X$.

Conversely, let $f \in X$ span an $N$-stable subspace of dimension 1. Thus $N$ acts on $f$ by right translation as a character. The support of $f$ is left-invariant under $B$, so $\operatorname{supp} f$ is either $G$ or $BwN$. The second case is impossible since if $f(1) = 0$, by the restriction induction lemma (theorem 2.3.4) the support of $f$ is confined to $BwN_0$, for some compact open subgroup $N_0$ of $N$. Thus, $\operatorname{supp} f = G$ and $f$ vanishes nowhere so that $f(1) \neq 0$, and $f \notin V$. Then the canonical $N$-map $X \to \mathbb{C} = X/V$ identifies the $N$-space $\mathbb{C}f$ with the trivial $N$-space $\mathbb{C}$. It follows that $N$ fixes $f$ under right translation. Take $x \in F^\times$, and consider the identity

$$w\begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & x^{-1} \\ 0 & 1 \end{pmatrix}\begin{pmatrix} -x^{-1} & 0 \\ 0 & x \end{pmatrix}\begin{pmatrix} 1 & 0 \\ x^{-1} & 1 \end{pmatrix}.$$

Taking $x$ so that $\|x\|$ is sufficiently large, then $f$ is fixed under right translation by $\begin{pmatrix} 1 & 0 \\ x^{-1} & 1 \end{pmatrix}$. So, as $f$ is fixed by $N$, we have

$$f(w) = \chi_1(-1)\chi_1^{-1}(x)\chi_2(x)f(1),$$

for all $x \in F^\times$ of sufficiently large absolute value. Thus $\chi_1 = \chi_2 = \varphi$, say, and $f(g) = \varphi(\det g)f(1)$. We have proved (1) $\Leftrightarrow$ (2), and that the one-dimensional $N$-subspace is uniquely determined.

We now prove theorem 2.3.8:

***Proof*:**
(of theorem 2.3.8)

Assume $X$ is reducible. By corollary 2.3.11 its $G$-length is 2 or 3, and it has either a finite-dimensional $G$-subspace or a finite-dimensional $G$-quotient

Assume the first alternative. Thus $X$ has a one-dimensional $N$-subspace, and we are in the situation of proposition 2.3.12: $X$ has a one-dimensional $G$-subspace $L$, and $\chi_1 = \chi_2 = \varphi$, say. Moreover, $G$ acts on $L$ as the character $\varphi \circ \det$ and $L \cap V = 0$. The quotient $Y = X/L$ is therefore $B$-isomorphic to $V$. If $X$ has $G$-length 3, then $Y$ has $G$-length 2. However, $V$ has $B$-length 2 and a unique $B$-quotient, which is of dimension 1. This gives a $G$-quotient of $Y$ on which $G$ must act as a character $\varphi' \circ \det$. This would force $\varphi' \otimes \varphi'$ to appear as a factor in the Jacquet module $Y_N \cong \varphi \cdot \delta_B^{-1}$, which it cannot. Thus $X$ has $G$-length 2 and we are in case (2)(c) of the Irreducibility Criterion (theorem 2.3.8).

In the other alternative, $X$ has a finite-dimensional $G$-quotient. The representation $\check{X}$ therefore has a finite-dimensional $G$-subspace, and we are back with the first alternative. By the Duality Theorem (theorem 1.3.12), $\check{X} \cong \mathrm{Ind}_B^G \delta_B^{-1}\check{\chi}$, so we are in the case (2)(d)of the Irreducibility Criterion (theorem 2.3.8).

Thus, in part (1) of the theorem, we have shown that $X$ reducible implies $\chi$ has the stated form. The converse is given by proposition 2.3.12 and the dual case. We have also proved statements (a)–(d).

Having fully determined how to get irreducible representations from inductions/inflation of characters, to get a classification of the irreducible (non-cuspidal) representations of $G$, we need to investigate the homomorphisms between induced representations:

**Proposition 2.3.13: Homomorphism between induced Representations**

Let $\chi, \xi$ be characters of $T$. The space $\mathrm{Hom}_G(\mathrm{Ind}_B^G \chi, \mathrm{Ind}_B^G \xi)$ has dimension 1 if $\xi = \chi$ or $\chi^w \delta_B^{-1}$, zero otherwise.

***Proof*:**
We use Frobenius Reciprocity:

$$\mathrm{Hom}_G(\mathrm{Ind}_B^G \chi, \mathrm{Ind}_B^G \xi) \cong \mathrm{Hom}_T((\mathrm{Ind}\, \chi)_N, \xi).$$

The Jacquet module $(\mathrm{Ind}\, \chi)_N$ fits into the exact sequence

$$0 \to \chi^w \delta_B^{-1} \to (\mathrm{Ind}\, \chi)_N \to \chi \to 0.$$

If we assume that $\chi \neq \chi^w \delta_B^{-1}$, then this sequence splits and the result follows immediately. The equation $\chi = \chi^w \delta_B^{-1}$ amounts to $\chi_1(x) = \|x\|\chi_2(x)$, $x \in F^\times$. In this case, $\mathrm{Ind}\, \chi$ is irreducible and the result again follows.

By way of some examples, we examine in more detail the case where $\mathrm{Ind}_B^G \chi$ is reducible. Thus there is a character $\varphi$ of $F^\times$ such that $\chi = \varphi \cdot 1_T$ or $\chi = \varphi \cdot \delta_B^{-1}$. Twisting does not affect the situation materially, so we assume $\varphi = 1$. Consider first the case $\mathrm{Ind}_B^G 1_T$. The irreducible $G$-quotient of

$\operatorname{Ind}_B^G 1_T$ is called the *Steinberg representation* of $G$, and is denoted $\mathrm{St}_G$:

$$0 \to 1_G \to \operatorname{Ind}_B^G 1_T \to \mathrm{St}_G \to 0. \tag{9.10.3}$$

Its dimension is infinite and $(\mathrm{St}_G)_N \cong \delta_B^{-1}$. Likewise, if $\varphi$ is a character of $F^\times$, we have an exact sequence

$$0 \to \varphi_G \to \operatorname{Ind}_B^G \varphi_T \to \varphi \cdot \mathrm{St}_G \to 0,$$

where $\varphi_G = \varphi \circ \det$ and $\varphi_T = \varphi \otimes \varphi$. (Representations of the form $\varphi \cdot \mathrm{St}_G$ are sometimes called *special*.)

Taking the smooth dual of (9.10.3), we get an exact sequence

$$0 \to \mathrm{St}_G^\vee \to \operatorname{Ind}_B^G \delta_B^{-1} \to 1_G \to 0. \tag{9.10.4}$$

The proposition implies

$$\mathrm{St}_G \cong \mathrm{St}_G^\vee. \tag{9.10.5}$$

*Remark.* The proposition also implies that the space $\operatorname{End}_G(\operatorname{Ind} 1_T)$ has dimension 1, while $\operatorname{Ind} 1_T$ is not irreducible. Thus the converse of Schur's Lemma fails in this context, as remarked in 2.6.

Observe also the imperfect parallelism between the proposition above and the corresponding result (6.3) for the finite field case.

9.11. We introduce a new notation. If $\sigma$ is a smooth representation of $T$, we define

$$\iota_B^G \sigma = \operatorname{Ind}_B^G(\delta_B^{-1/2} \otimes \sigma). \tag{9.11.1}$$

This provides another exact functor $\mathrm{Rep}(T) \to \mathrm{Rep}(G)$, known as *normalized* or *unitary smooth induction*. It gives rise to more convenient combinatorics, for example:

$$(\iota_B^G \sigma)^\vee \cong \iota_B^G \check{\sigma}. \tag{9.11.2}$$

In this language, the Irreducibility Criterion (9.6) and 9.10 Proposition say:

**Lemma 2.3.14**

(1) Let $\chi = \chi_1 \otimes \chi_2$ be a character of $T$. The representation $\iota_B^G \chi$ is reducible if and only if $\chi_1 \chi_2^{-1}$ is one of the characters $x \mapsto \|x\|^{\pm 1}$ of $F^\times$ or, equivalently, $\chi = \varphi \cdot \delta_B^{\pm 1/2}$ for some character $\varphi$ of $F^\times$. (2) Let $\chi, \xi$ be characters of $T$. The space $\operatorname{Hom}_G(\iota_B^G \chi, \iota_B^G \xi)$ is not zero if and only if $\xi = \chi$ or $\xi = \chi^w$.

Gathering up our earlier arguments and results, we get:

**Theorem 2.3.15: Classification Theorem**

The following is a complete list of the isomorphism classes of irreducible, non-cuspidal representations of $G$: (1) the irreducible induced representations $\iota_B^G \chi$, where $\chi \neq \varphi \cdot \delta_B^{\pm 1/2}$ for any character $\varphi$ of $F^\times$; (2) the one-dimensional representations $\varphi \circ \det$, where $\varphi$ ranges over the characters of $F^\times$; (3) the special representations $\varphi \cdot \mathrm{St}_G$, where $\varphi$ ranges over the characters of $F^\times$.
The classes in this list are all distinct except that, in (1), we have $\iota_B^G \chi \cong \iota_B^G \chi^w$.

***Proof*:**
The examples in 9.10 show that every irreducible, non-cuspidal representation of $G$ appears in this list. The relations between the irreducibly induced ones are given by 9.10 Proposition. The same result also implies that no special representation $\varphi \cdot \mathrm{St}_G$ can be equivalent to $\varphi' \cdot \mathrm{St}_G$, $\varphi' \neq \varphi$, or any irreducibly induced representation.

## 2.4 Cuspidal Representations and Coefficients

(not sure if I want to do with this yet, but here until I absorb the material)

Our exploration of non-principal series representations interestingly enough starts revisiting the definition of the cuspidal, so far defined as a smooth representation $(\pi, V)$ of $\mathrm{GL}_2(F)$ where the Jacquet module $(\pi_N, V_N)$ is trivial, equivalently, where it is not isomorphic to a composition factor of an induced representation $\mathrm{Ind}_B^G \chi$, for any character $\chi$ of $T$. This definition of being the "negative" of non-principal series hides the great importance these have in Langlands and hence we start with giving a more practical and algebraic definition.

The key insight from the soon-to-be-shown new viewpoint is that irreducible cuspidal representations have an algebraic property: they are *projective objects* in the appropriate subcategory of $\mathrm{Rep}(G)$, as shall be elaborated on in ref:HERE (10a in [2]).

Let $(\pi, V)$ be a smooth representation of $G = \mathrm{GL}_2(F)$; from vectors $v \in V$, $\tilde{v} \in \check{V}$, we get a smooth function on $G$ by

$$\gamma_{\tilde{v}\otimes v} : g \mapsto \langle \tilde{v}, \pi(g)v \rangle.$$

Let $C(\pi)$ be the vector space spanned by the functions $\gamma_{\tilde{v}\otimes v}$, $\tilde{v} \in \check{V}, v \in V$. The functions $f \in C(\pi)$ are called the *(matrix) coefficients* of $\pi$. The space $\check{V} \otimes V$ carries a smooth representation of the group $G \times G$, while $G \times G$ acts on the function space $C(\pi)$ by translation: the first factor acts by left translation and the second by right translation. The map $\tilde{v} \otimes v \mapsto \gamma_{\tilde{v}\otimes v}$ is then a surjective $G \times G$-homomorphism $\check{V} \otimes V \to C(\pi)$.

The space $C(\pi)$ is primarily, but not exclusively, of interest in the case where $\pi$ is irreducible. When $\pi$ is irreducible, the centre $Z$ of $G$ acts on $V$ via the central character $\omega_\pi$ of $\pi$ and

$$\gamma(zg) = \omega_\pi(z)\gamma(g), \quad z \in Z, g \in G, \gamma \in C(\pi).$$

The support of a coefficient is therefore invariant under translation by $Z$.

**Definition 2.4.1: Cuspidal Representation**

Let $(\pi, V)$ be an irreducible smooth representation of $G$; one says that $\pi$ is $\gamma$-cuspidal if every $\gamma \in C(\pi)$ is compactly supported modulo $Z$.

The term "$\gamma$-cuspidal" is a convenient, but temporary. If unambiguous, a representation is described as cuspidal or $\gamma$-cuspidal, it is implicitly assumed to be smooth.

# 3

# *Cuspidal Representations*

here (I think I'll also combine chapter 5 of the book here)

# 4

# *Characterizing Representations using Functional Equations*

(This is important as the key *Converse Theorem* is key to the Langlands Correspondence)

# 5

# Representation of Galois Theory: Weil Groups

(here; will do a write-up once I understand more).

## 5.1 Weil Group and Smooth Representations

We start by recalling some features of the Galois theory of $F$. Throughout, let $p$ means the characteristic of the residue class field $\mathbf{k} = \mathfrak{o}/\mathfrak{p}$. Choose a separable algebraic closure $\overline{F}$ of $F$. We set $\Omega_F = \mathrm{Gal}(\overline{F}/F)$, and view it as a profinite group with its natural (Krull) topology. Thus

$$\Omega_F = \varprojlim \mathrm{Gal}(E/F),$$

where $E/F$ ranges over finite Galois extensions with $E \subset \overline{F}$. If $K/F$ is a finite field extension, with $K \subset \overline{F}$, denote by $\Omega_K$ the open subgroup $\mathrm{Gal}(\overline{F}/K)$ of $\Omega_F$. If $\widetilde{F}/F$ is some other separable algebraic closure of $F$, an $F$-isomorphism $\widetilde{F} \cong \overline{F}$ induces an isomorphism $\mathrm{Gal}(\widetilde{F}/F) \cong \Omega_F$. This does depend on the choice of isomorphism $\widetilde{F} \cong \overline{F}$. Fortunately, it is only up to inner automorphism, so in particular, it induces a *canonical* bijection between the sets of equivalence classes of irreducible smooth representations of the two groups $\mathrm{Gal}(\widetilde{F}/F)$, $\Omega_F$.

Fixing an $\overline{F}/F$, consider $E/F$ where $E \subset \overline{F}$. Then the field $F$ admits a unique unramified extension $F_m/F$ of degree $m$, for each $m \geqslant 1$. Let $F_\infty$ denote the composite of all these fields. Then $F_\infty/F$ is the unique maximal unramified extension of $F$. The extension $F_m/F$ is Galois and $\mathrm{Gal}(F_m/F)$ is cyclic. An $F$-automorphism of $F_m$ is determined by its action on the residue field $\mathbf{k}_{F_m} \cong \mathbb{F}_{q^m}$ (see [5, section 1.6]). In particular, there is a unique element $\varphi_m$ of $\mathrm{Gal}(F_m/F)$ which acts on $\mathbf{k}_{F_m}$ as $x \mapsto x^q$. Set $\Phi_m$ to be the map for $\mathrm{Gal}(F_m/F)$ corresponding to the Frobenius map $\varphi_m^{-1}$ for the extension of finite fields. The map $\Phi_m \mapsto 1$ gives a canonical isomorphism $\mathrm{Gal}(F_m/F) \to \mathbb{Z}/m\mathbb{Z}$. Taking the limit over $m$, we get a canonical isomorphism of topological groups

$$\mathrm{Gal}(F_\infty/F) \cong \varprojlim \mathbb{Z}/m\mathbb{Z} = \widehat{\mathbb{Z}}$$

and a unique element $\Phi_F \in \mathrm{Gal}(F_\infty/F)$ which acts on $F_m$ as $\Phi_m$, for all $m$. Note that $\langle \Phi_F \rangle \mathbb{Z}$ and hence cannot generate the entire $\mathrm{Gal}(F_\infty/F)$, however it is *dense* (on every finite group the restriction generates the entire group).

**Definition 5.1.1: Geometric Frobenius**

The element $\Phi_F$ is the *geometric Frobenius substitution* on $F_\infty$. The map $\varphi_F$ for the limit of galois groups of finite fields is the *arithmetic Frobenius substitution.*

An element of $\Omega_F$ is called a *geometric Frobenius element* (over $F$) if its image in $\mathrm{Gal}(F_\infty/F)$ is $\Phi_F$. The Chinese Remainder Theorem gives a canonical isomorphism of topological groups

$$\widehat{\mathbb{Z}} \cong \prod_\ell \mathbb{Z}_\ell,$$

where $\ell$ ranges over all prime numbers and $\mathbb{Z}_\ell$ is the (additive) group of $\ell$-adic integers.

Let $\mathcal{I}_F = \mathrm{Gal}(\overline{F}/F_\infty)$ be the *inertia group* of $F$. This fits in the exact sequence of topological groups

$$1 \to \mathcal{I}_F \to \Omega_F \to \widehat{\mathbb{Z}} \to 0.$$

Let us now extend to the maximal tamely ramified extensions.For each integer $n \geqslant 1$, $p \nmid n$, the field $F_\infty$ has a unique extension $E_n/F_\infty$ of degree $n$. If $\varpi$ is a prime element of $F$, the extension $E_n$ is generated over $F_\infty$ by an $n$-th root of $\varpi$. The extension $E_n/F_\infty$ is cyclic, and the Galois group $\mathrm{Gal}(E_n/F_\infty)$ admits a canonical description using Kummer theory (presented in [6, section 18.4]). Choose $\alpha \in E_n$ such that $\alpha^n = \varpi$. The map

$$\begin{aligned} \mathrm{Gal}(E_n/F_\infty) &\longrightarrow \mu_n, \\ \sigma &\longmapsto \sigma(\alpha)/\alpha, \end{aligned} \tag{5.1}$$

gives a canonical isomorphism of $\mathrm{Gal}(E_n/F_\infty)$ with the group $\mu_n$ of $n$-th roots of unity in $F_\infty$. The composite $E_\infty$ of all extensions $E_n/F_\infty$ is the maximal tamely ramified extension of $F$. Taking the limit of the isomorphisms (28.3.1), we get topological isomorphisms

$$\mathrm{Gal}(E_\infty/F_\infty) \cong \varprojlim \mu_n \cong \prod_{\ell \neq p} \mathbb{Z}_\ell.$$

In the limit, $n$ ranges over all positive integers not divisible by $p$; in the product, $\ell$ ranges over all prime numbers other than $p$.

Finally, let $\mathcal{P}_F = \mathrm{Gal}(\overline{F}/E_\infty)$ be the *wild inertia group*, completing the tower

$$\overline{F}/E_\infty/F_\infty/F$$

Any finite extension of $E_\infty$ has degree a power of $p$, so the profinite group $\mathcal{P}_F$ is a pro $p$-group. It is the unique pro $p$-Sylow subgroup of $\mathcal{I}_F$, and hence it is normal, giving the isomorphism $\mathcal{I}_F/\mathcal{P}_F \cong \prod_{\ell \neq p} \mathbb{Z}_\ell$ which is uniquely determined up to multiplication by an element of $\prod \mathbb{Z}_\ell^\times$.

The group $\mathrm{Gal}(F_\infty/F)$ acts on $\mathrm{Gal}(E_\infty/F_\infty)$ by conjugation. An isomorphism transfers this action to one on the group $\prod_{\ell\neq p}\mathbb{Z}_\ell$. For future reference, we box this result:

**Proposition 5.1.2: Tame Inertia Action**

If $t_0 : \mathcal{I}_F/\mathcal{P}_F \to \prod_{\ell\neq p}\mathbb{Z}_\ell$ is a topological isomorphism, then

$$t_0(\Phi_F\sigma\Phi_F^{-1}) = q^{-1}t_0(\sigma), \quad \sigma \in \mathcal{I}_F/\mathcal{P}_F.$$

Let ${}_a\mathcal{W}_F$ be defined by:

$${}_a\mathcal{W}_F = \{\sigma \in \Omega_F \ : \ \sigma|_{F_\infty} \in \langle\Phi_F\rangle\} = \mathrm{res}^{-1}(\langle\Phi_F\rangle) \subseteq \Omega_F,$$

where $\mathrm{res} : \Omega_F \to \mathrm{Gal}(F_\infty/F)$ is the restriction map. As any $\varphi \in \mathcal{I}_F$ restricts to the identity, $\mathcal{I}_F \subseteq {}_a\mathcal{W}_F$. For If $G_E = \mathrm{Gal}(E/F)$ for a finite $F$, Then the restriction to $E$ is surjective, and hence ${}_a\mathcal{W}_F$ is *dense* in $\mathrm{Gal}(\overline{F}/F)$.

Next, the map:

$$\mathrm{res} : {}_a\mathcal{W}_F \to \mathrm{Gal}(F_\infty/F) \qquad \sigma \mapsto \sigma|_{F_\infty}$$

is by construction surjective onto $\langle\Phi_F\rangle$ with kernel $\mathcal{I}_F$ forming a short exact sequence:

$$1 \to \mathcal{I}_F \to {}_a\mathcal{W}_F \to \mathbb{Z} \to 0$$

Notice the difference between these two quotients:

$${}_a\mathcal{W}_F/\mathcal{I}_F \cong \mathbb{Z} \quad \text{(Generated by Frobenius)} \qquad \text{and} \qquad \mathrm{Gal}(F_\infty/F) \cong \widehat{\mathbb{Z}} \quad \text{(Profinite completion)}$$

| Inertia (Vertical) | | Total Group | | Unramified (Horizontal) |
|---|---|---|---|---|
| $\mathcal{I}_F$ | $\subset$ | $\Omega_F$ (Galois) | $\twoheadrightarrow$ | $\widehat{\mathbb{Z}}$ |
| $\vert$ | | $\cup$ | | $\cup$ |
| $\mathcal{I}_F$ | $\subset$ | ${}_a\mathcal{W}_F$ (Weil) | $\twoheadrightarrow$ | $\mathbb{Z}$ |

For a concrete element in $\Omega_F$ that is not in ${}_a\mathcal{W}_F$, assume $\mathrm{char}F \neq 2$ so that 2 is invertible in $F$ (or else pick 3). Let $\alpha$ be the element that is $1/2 \bmod p^n$ for each $n$ and $p \neq 2$. Let $\sigma \in \Omega_F$ associate to this element and notice that $\sigma^2|_{F_\infty} = \Phi_F$, showing that it cannot be in ${}_a\mathcal{W}_F$.

Now, ${}_a\mathcal{W}_F$ is a profinite group, hence not "enough" representations to correspond to irreducible representations mentioned earlier. We need a better topology to hold the arithmetic information of ${}_a\mathcal{W}_F$, that is a locally profinite topology:

**Definition 5.1.3: Weil Group**

The Weil group $\mathcal{W}_F$ of $F$ (relative to $\bar{F}/F$) is the topological group, with underlying abstract group $a\mathcal{W}_F$, so that

1. $\mathcal{I}_F$ is an open subgroup of $\mathcal{W}_F$, and
2. the topology on $\mathcal{I}_F$, as subspace of $\mathcal{W}_F$, coincides with its natural topology as $\mathrm{Gal}(\bar{F}/F_\infty) \subset \Omega_F$.

As $\mathcal{I}_F$ is open, $\mathcal{W}_F/\mathcal{I}_F \cong \mathbb{Z}$ must be discrete, and as it is infinite it must be that $\mathcal{W}_F$ is no longer compact, and hence it is *locally profinite* (recall in a profinite group a set is open if and only if it is finite index). The definition of $\mathcal{W}_F$ does depend on the choice of $\overline{F}/F$ up to inner automorphism of $\Omega_F$.

**Lemma 5.1.4: Injection is Continuous**

The identity map $\iota_F : \mathcal{W}_F \to {}_a\mathcal{W}_F \subset \Omega_F$ is a continuous injection.

***Proof*:**
The topology of $\mathcal{W}_F$ is strictly finer.

Let $\nu_F : \mathcal{W}_F \to \mathbb{Z}$ be the canonical map taking a geometric Frobenius element to 1 and define:

$$\|x\| = q^{-v_F(x)}, \quad x \in \mathcal{W}_F.$$

**Proposition 5.1.5: Weil subgroup and Finite Extensions**

1. Let $E/F$ be a finite extension, $E \subset \overline{F}$.
   (a) The group $\mathcal{W}_F$ has a unique subgroup $\mathcal{W}_F^E$ such that
   $$\iota_F(\mathcal{W}_F^E) = {}_a\mathcal{W}_F \cap \Omega_E.$$
   (b) The subgroup $\mathcal{W}_F^E$ is open and of finite index in $\mathcal{W}_F$; it is normal in $\mathcal{W}_F$ if and only if $E/F$ is Galois.
   (c) The canonical map $\mathcal{W}_F^E\backslash\mathcal{W}_F \to \Omega_E\backslash\Omega_F$ is a bijection.
   (d) The canonical map $\iota_E : \mathcal{W}_E \to \Omega_E$ induces a topological isomorphism $\mathcal{W}_E \cong \mathcal{W}_F^E$.
2. The map $E/F \mapsto \mathcal{W}_F^E$ is a bijection between the set of finite extensions $E$ of $F$ inside $\overline{F}$ and the set of open subgroups of $\mathcal{W}_F$ of finite index.

***Proof*:**
exercise.

We henceforward, identify $\mathcal{W}_E$ with the subgroup $\mathcal{W}_F^E$ of $\mathcal{W}_F$.

## (Complex) Smooth Representation of Weil Group

We now look at smooth representations of $\mathcal{W}_F$. Note that as $\Omega_F$ is profinite, smooth representations must be semi-simple, while $\mathcal{W}_F$ being locally profinite (as it admits a discrete quotient $\mathbb{Z}$) must admit non-semisimple representations. As we will see, the irreducible representations of $\mathcal{W}_F$ are closely related to the irreducible representations of $\Omega_F$ , but the reducible ones exhibit a greater variety, namely the principal series shall be associated to them, and these are "hidden" if we only look at $\Omega_F$. Just like in the previous chapters, we focus on *complex* representations. P-adic representations are left for ref:HERE.

**Lemma 5.1.6: Irreducible Representations of Weil Group is Finite**

Let $(\rho, V)$ be an irreducible representation of $\mathcal{W}_F$. Then $V$ is finite dimensional.

***Proof*:**
let $v \in V$. As $\rho$ is smooth it is fixed by a compact open subgroup $K \subseteq \mathcal{W}_F$. Let $\mathcal{J} = K \cap \mathcal{I}_F$ so that $v$ is fixed by an open subgroup of $\mathcal{I}_F$. As $\mathcal{I}_F$ has the same topology as the subspace topology of $\mathcal{I}_F \subseteq \Omega_F$ and $\mathcal{J}$ is open, it corresponds to a finite extension $E/F$. By taking the galois closure of $E/F$, we get a smaller group that is still in the stabilizer of $v$, hence we may assume $E/F$ is galois hence $\mathcal{J}$ is normal in $\mathcal{W}_F$. It is an exercise in classical representation theory to show that if $\rho$ is irreducible and $N \trianglelefteq G$ is normal, if $N$ fixes a point $v$, $N$ acts trivially on all of $V$. Hence, $\mathcal{J} \subseteq \ker \rho$.

Now, as $\mathcal{J} \subseteq \mathcal{I}_K$ is an open subgroup, $\mathcal{I}_K/\mathcal{J}$ is finite. Let $\Phi \in \mathcal{W}_F$ be a Frobenius element (an element of the coset $1 + \mathcal{I}_F$ in the map $\mathcal{W}_F \xrightarrow{\text{res}} \langle \Phi_F \rangle \cong \mathbb{Z}$). As both $\mathcal{I}_K$ and $\mathcal{J}$ are normal in $\mathcal{W}_F$ (they are both associated to a galois group of an extension of $F$), $\Phi$ acts by conjugation on $\mathcal{I}_K/\mathcal{J}$, and as this group is finite some $\Phi^d$ acts trivially on this group. Thus:

$$\Phi^d \cdot x \cdot \Phi^{-d} = xj$$

for appropriate $j \in \mathcal{J}$. Applying $\rho$:

$$\rho(\Phi^d)\rho(x)\rho(\Phi^{-d}) = \rho(x)\rho(j)$$

As $\mathcal{J} \subseteq \ker \rho$:

$$\rho(\Phi^d)\rho(x)\rho(\Phi^{-d}) = \rho(x) \cdot I = \rho(x)$$

Moving terms around:

$$\rho(\Phi^d)\rho(x) = \rho(x)\rho(\Phi^d)$$

Thus the map $\rho(\Phi^d)$ is in the center of $\rho(\mathcal{W}_F) \subseteq \mathrm{Aut}(V)$, and as $\rho$ is irreducible this means $\rho(\Phi^d)$ acts like scalar multiplication.

Using this, I claim that for the $v$ that was fixed at the very beginning of the proof, the space

$$U = \{\rho(g) \cdot v \ : \ g \in \mathcal{W}_F\}$$

is finite dimensional. By definition of $\mathcal{W}_F$, $g = \Phi^k t$ for $t \in \mathcal{I}_F$. By taking the finite $\mathcal{I}_F/\mathcal{J}$ we have coset representatives $r_1, ..., r_n$ so that:

$$\rho(g) \cdot v = \rho(\Phi^k)\rho(r_\alpha)\rho(j) \cdot v$$

where $j \in \mathcal{J}$ is any element. As $\mathcal{J} \subseteq \ker \rho$ we simplify to

$$\rho(g) \cdot v = \rho(\Phi^k)\rho(r_\alpha) \cdot v$$

Next, write $\Phi^k = \Phi^{ad+c}$ where $0 \leq c < d$ so that we can use the fact that $\rho(\Phi^d)$ is scalar multiplication to get

$$\rho(\Phi^d) = \lambda^a \rho(\Phi^c)$$

But now, we get that there finitely many possible vectors that are used to span $U$

$$\rho(\Phi^c)\rho(r_\alpha) \cdot v \qquad 0 \leq c < d, \text{ and } 0 < \alpha < n$$

showing it is finite. Certainly this space is $\mathcal{W}_F$-stable, and as $V$ is irreducible it must be that $U = V$, as we sought to show.

Hence the representations of $\mathcal{W}_F$ are interest will be finite! Let $\rho$ be a finite representation of $\mathcal{W}_F$. Then the restriction to $\mathcal{I}_F$ (a profinite group) will be semi-simple by an easy generalization of Maschke's theorem (proposition 1.2.9), and $\rho(\mathcal{I}_F)$ is finite. Therefore, all the complexity comes from the Frobenius elements of $\mathcal{W}_F$.

# 6

# *Langlands Correspondence*

(Reaching here is the goal of this book)

# Index

# *Bibliography*


[1] Nathanael Chwojko-Srawley. *Everything You Need To Know About Galois Cohomology.* EYNTKA. 2025. URL: https://nathanaelsrawley.com/assets/pdfs/books/EYNTKA_galois_cohomology.pdf.

[2] Colin J. Bushnell and Guy Henniart. *The local Langlands conjecture for GL(2).* Grundlehren der mathematischen Wissenschaften. Springer, 2006. ISBN: 978-3-540-31486-8.

[3] Nathanael Chwojko-Srawley. *Everything You Need To Know About Functional Analysis.* 1st ed. NA. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_functional_analysis.pdf.

[4] Nathanael Chwojko-Srawley. *Everything You Need To Know About Real Analysis.* 1st ed. Vol. 1. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_real_analysis.pdf.

[5] Nathanael Chwojko-Srawley. *Everything You Need To Know About Class Field Theory.* URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_class_field_theory.pdf.

[6] Nathanael Chwojko-Srawley. *Everything You Need To Know About Undergraduate Algebra.* 1st ed. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_algebra.pdf.